Universidad del Valle

Trabajo de Grado

Implementación de un método numérico basado en redes tensoriales para el análisis de un sistema reticular de bosones de espín-1 bajo influencia de un campo magnético de Zeeman

Implementation of a numerical method based on tensor networks for the analysis of a lattice system of spin-1 bosons under the influence of a Zeeman magnetic field

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To my parents

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0.1 Publications

The following is a list of publications produced during the development of this project:

[1] J. S. Gómez and K. C. Rodríguez, “Bibliometric Analysis and Overview of Matrix Product States in the Bose-Hubbard Model,” *Ingeniería*, vol. 30, no. 1, p. e22292, Mar. 2025. https://doi.org/10.14483/23448393.22292

Resumen

Desde la predicción de Bose y Einstein en 1925 hasta el hito experimental en 1995 con la realización en laboratorio del condensado de Bose-Einstein, los gases ultrafríos han evolucionado para convertirse en simuladores cuánticos altamente controlables. Campos ópticos de ondas estacionarias, las redes ópticas, ofrecen un análogo libre de defectos a un cristal donde las interacciones, el llenado y la dimensionalidad se pueden ajustar con precisión experimental. En este escenario le surgió a Hubbard la idea original para electrones, la cual fue adaptada a bosones y posteriormente extendida para incluir el espín interno, habilitando colisiones con cambio de espín y control de Zeeman.

Esta tesis se inscribe en esa tradición, estudiamos el modelo de Bose-Hubbard de espín-1 en 1D bajo un campo de Zeeman lineal, un régimen donde las fluctuaciones son fuertes y tanto los experimentos como los métodos numéricos robustos resultan esenciales. En el frente computacional, la trayectoria va desde el método del Grupo de Renormalización de Matriz de Densidad (DMRG) hasta su formulación moderna en términos de Estados de Producto de Matrices (MPS), redes tensoriales que aprovechan la ley de área del entrelazamiento para obtener estados fundamentales precisos en 1D. Siguiendo esa línea, y con el objetivo de mantener la máquina transparente para la física de spin-1, implementamos desde cero una librería en C++ de MPS/MPO, una base nativa en la proyección magnética, y protocolos eficientes para la evaluación de observables locales y correladores de dos puntos. Una traza de “matriz de selección” basada en la matriz de transferencia permite comprimir flujos únicamente diagonales y reducir la carga computacional.

Una contribución distintiva de este trabajo es el nivel de detalle dedicado al formalismo de los Operadores de Producto de Matrices (MPO). La implementación se presenta con protocolos explícitos para la contracción de tensores sobre la Matriz de Transferencia extendida para incluir la dimensión del enlace del MPO, fielmente descritos y esquematizados para mostrar cómo se optimiza el rendimiento en el núcleo del código. Este grado de transparencia, poco común en la literatura, ofrece tanto una guía práctica de implementación como un mapa conceptual del funcionamiento interno de las rutinas basadas en MPO.

El análisis numérico se desarrolla en dos etapas. Se presenta una comparación uno a uno entre los valores de expectativa obtenidos con el código variacional previo basado en MPS y los calculados mediante la nueva implementación con MPO, demostrando una excelente concordancia y confirmando la validez del nuevo desarrollo. Una vez validados los protocolos, caracterizamos los estados de manera exclusiva dentro del marco de MPO. Esta comparación revela cómo los operadores no locales son tratados de forma fundamentalmente distinta tanto en MPS como en MPO, siendo

este último el que permite tanto mejoras en programación de alto rendimiento como una mayor versatilidad para futuros modelos físicos.

Finalmente, validamos los protocolos basados en MPO frente a rutinas anteriores de MPS y los aplicamos en los regímenes aislante de Mott y superfluido del modelo de Bose-Hubbard, siguiendo la evolución de observables locales, longitudes de correlación y desplazamientos de banda inducidos por el campo lineal de Zeeman; en el límite de acoplamiento fuerte recuperamos patrones característicos del régimen Mott (sectores tipo Dimer, Large-D y XY ferromagnético) consistentes con el panorama histórico del modelo. En suma, el presente trabajo revisita un modelo de Bose-Hubbard enriquecido con interacciones espinor no triviales y ajuste Zeeman, al mismo tiempo que contribuye con una caja de herramientas tensoriales de código abierto diseñada para su extensibilidad hacia cálculos con DMRG/iTEBD e iMPS, ampliando así el conjunto de herramientas computacionales de nuestro grupo de Física del Estado Sólido Teórica en la Universidad del Valle.

Palabras Clave: Modelo de Bose-Hubbard, bosones de spin-1, redes ópticas,campo linear de Zeeman, Aislante de Mott, Fase superfluida, transición de fases cuántica, sistemas cuántico de muchos cuerpos, redes tensoriales, estado producto de matrices, operador producto de matrices, matriz de transferencia.

Abstract

From Bose and Einstein's 1925 prediction to the 1995 laboratory realization of Bose-Einstein condensation, ultracold gases have evolved into tunable quantum simulators. Standing-wave light fields, optical lattices, provide defect-free crystal analogs where interactions, filling, and dimensionality are dialed with experimental control. Within this setting, the Hubbard idea, originally devised for electrons, was repurposed for bosons and then extended to internal spin, enabling spin-changing collisions and Zeeman control.

This thesis goes on that track, we study a one-dimensional spin-1 Bose-Hubbard model under a linear Zeeman field, a regime where fluctuations are strongly-controlled experiments and robust numerics matter. On the computational side, the trajectory runs from DMRG to its modern Matrix Product State (MPS) formulation: tensor networks that exploit area-law entanglement to deliver accurate ground states in 1D. Following that tradition, and to keep the machinery transparent for spinor physics, we implement from scratch a C++ MPS/MPO library, a native magnetic-projection basis, and efficient evaluation of local observables and two-point correlators. A transfer-matrix "selection-matrix" trace compresses diagonal-only flows and reduces computational overhead.

A distinctive contribution of this work lies in the unusually detailed treatment of the Matrix Product Operator formalism. The implementation is presented with explicit protocols for tensor contraction on the extended Transfer Matrix, faithfully described and schematized to illustrate how performance is optimized within the core of the code. This level of transparency, rarely available in the literature, offers both a practical guide for implementation and a conceptual map of the internal workings of MPO-based routines.

The numerical analysis proceeds in two stages. A one-to-one comparison between expectation values computed with the previous MPS-variational code and those obtained through the new MPO-based implementation demonstrates excellent agreement, confirming the soundness of the newly implemented approach. Having validated the protocols, we then characterize the states exclusively within the MPO framework. This comparison reveals how non-local operators are treated in fundamentally different ways by MPS and MPO methods, with the latter enabling both high-performance upgrades and greater versatility for prospective physical models.

We validate the MPO protocols against prior MPS routines and apply them across Mott-Insulating and Superfluid regimes of the Bose-Hubbard model, tracking local quantities, correlation lengths, and Zeeman-driven band shifts; in the strong-coupling limit we recover characteristic Mott-regime patterns (Dimer, Large-D, XY-Ferromagnetic like sectors) consistent with the historical model landscape. In sum, the present work revisits a nourished Bose-Hubbard modelwith non-trivial types of spinor interactions and existance of Zeeman-tunning, while also contribuiting an “open-hood” tensor-network toolkit designed for extensibility towards DMRG/iTEBD and iMPS calculations, increasing the suite of computational tools for our research group, the Theoretical Solid State Physics at the Universidad del Valle.

Key words: Bose–Hubbard model, spin-1 bosons, spinor gases, optical lattices, linear Zeeman field,mott-insulator, superfluid phase, quantum phase transition, quantum-many body systems, tensor networks, Matrix Product States, Matrix Product Operators, transfer matrix.

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Chapter 1

Introduction

One of the most remarkable milestones in 20th century quantum physics was the experimental realization of the Bose-Einstein condensate (BEC) in 1995, achieved independently by Cornell and Wieman at JILA (with ^{87}Rb) and by Ketterle at MIT (with ^{23}Na) [2, 3]. This breakthrough confirmed a phenomenon first predicted by Satyendra Nath Bose and Albert Einstein in 1925, a new state of matter in which, at ultra-low temperatures, a macroscopic fraction of particles collapses into the lowest energy quantum state [4, 5]. Such conditions became accessible thanks to laser cooling [6], magnetic trapping [7], and evaporative cooling techniques pioneered from the mid 1970s onward [8, 9].

Since the foundational statistical mechanical ideas of Bose and Einstein and their later experimental confirmation, ultracold quantum gases have become a vibrant arena where multiple branches of physics converge, from condensed matter to quantum information and quantum simulation [10]. The key to this progress is the control afforded by modern techniques, radiation pressure and dipole forces enable laser cooling and trapping of dilute atomic ensembles at microkelvin and nanokelvin scales, while standing wave light fields create *periodic potentials*, *optical lattices*, that emulate crystalline structures without defects or phonons [11]. Within this setting, strongly correlated bosons in lattices provide a clean platform to study emergent quantum phases, critical phenomena, and nonequilibrium dynamics.

Among lattice models with short range interactions, the Bose-Hubbard (BH) model occupies a central place. Introduced to capture tunneling and on-site repulsion of bosons on a discrete lattice [12], it has been scrutinized using mean-field and dynamical mean-field approaches [13, 14], diagrammatic techniques [15], exact diagonalization, and quantum Monte Carlo (QMC) [16]. Beyond spinless bosons, adding internal spin dramatically enriches the phase structure. For spin-1 bosons, magnetic and nematic order parameters can coexist with superfluidity, opening routes to study spin-changing collisions and symmetry breaking in a tunable setting [17].

This thesis focuses on the one-dimensional (1D) spin-1 BH model under an external Zeeman field, one dimension is both theoretically and experimentally compelling, enhanced quantum fluctuations suppress conventional long range order and promote Luttinger liquid behavior and phase transitions.

Spinor gases in optical lattices add controllable interactions through scattering channels with total added spin $S = \{0, 2\}$ (See Table A.1), while a linear Zeeman field B tunes the relative energies of magnetic sublevels. Present-day experiments can independently adjust lattice depth, interaction strengths, and magnetic fields, making the regime considered here directly relevant [11].

While the spinless BH model is textbook in 1D and higher dimensions, the spin-1 extension in *one dimension* is comparatively less mapped at high numerical accuracy. Much of the literature relies on mean field or higher dimensional analyses, or concentrates on special parameter regimes. Detailed Tensor Networks studies that operate directly in the spin projection basis relevant to experiments with spin dependent interactions are rare.This work focuses on that niche by introducing a measurement-centric Matrix Product State (MPS) framework, implemented through Matrix Product Operators (MPOs). The method is compatible with any computational basis, but is specifically tailored for spin-1 bosons with on-site contact interactions between boson pairs, resulting in local physical dimension of $d = 4 \times 4 = 16$, since each site hosts two interacting spin-1 bosons and the on-site Hilbert space is the tensor product of their individual four-dimensional spin subspaces.

A central challenge in many-body physics is the exponential growth of Hilbert space with system size. The density-matrix renormalization group (DMRG) [18] and its modern formulation in terms of MPS address this in 1D by targeting the low entanglement corner of Hilbert space. Tensor networks naturally encode area law entanglement, avoid sign problem limitations that can obstruct Quantum Monte Carlo (QMC) for some models, and offer controlled accuracy via the bond dimension. In this project we adopt a hands on approach. We implement from scratch in C++ an MPS codebase extended with MPOs for measurements (see section 3.5) and Hamiltonian representations, emphasizing transparent tensor reshaping and contraction schemes. This complements existing libraries by exposing the inner workings needed to tailor algorithms to spinor BH physics [19]. The implementation is designed to support periodic systems with translational symmetry, operate natively in the magnetic-projection basis, and provide efficient evaluation of local and two-point correlators (see section 4.1) relevant for phase characterization, including a selection-matrix strategy for traces.

Although several sections adopt a personal, literary style, the scientific content remains rigorous throughout arguments, models, and results are presented with full mathematical, numerical, and physical consistency. The aim is to make the exposition engaging without compromising precision.

This thesis is organized as follows. Chapter 2.2.1 introduces the physical system spin-1 bosons in a 1D optical lattice summarizing the Bose-Hubbard model, the role of the Zeeman field, and the scattering channels that shape the effective interactions, while fixing notation and conventions. Chapter 3 develops the Tensor Network (TN) framework used throughout, presenting MPS and MPO representations, transfer matrices, and canonical forms, together with the key implementation details needed for measurements. Chapter 4 compiles the main numerical results, contrasting MPO based evaluations with earlier MPS routines and analyzing local observables and two point correlators across representative parameter regimes. Finally, Chapter 5 summarizes the conclusions and delineates next steps grounded in the present library, inaugurating a new numerical track in our group, including time evolution methods and optimized network contractions.

Chapter 2

Spin-1 Systems in Optical Lattices

In condensed matter physics, a wide variety of systems can be explored. Our specific interest lies in those that have been promising candidates for quantum simulators[20, 21, 22, 23] in the last decade, which can be realized using cold atoms in optical lattices. Such systems offer a versatile platform with potential applications across multiple areas of physics, including quantum information theory[24], solid-state physics[25], and quantum computing[26]. In recent years, most quantum simulators have been based on variations of one of the simplest and most fundamental models in quantum many-body physics: the Hubbard model[16].

Before introducing the specific model we study, it is worth recalling that the original Hubbard model (on which our work is based) was first proposed to describe electrons in solid-state systems[12]. It served as a simplified "toy model" to investigate the magnetic properties of electrons in transition metals. Over time, the model has been extended by introducing additional terms, allowing it to describe a wide variety of systems beyond electrons, including bosons[27], fermions, and mixtures of both[28]. In our case, we focus on spin-1 bosons, whose additional spin degree of freedom allows for a richer variety of quantum phases and dynamical behavior compared to spin-$\frac{1}{2}$ systems.

2.1 Bose-Hubbard Model

To get started we have to remember that everything that's happening in this movie called our system takes place in scenes, to be precise, it's happening in a stage, which is a 1D optical lattice, that provides an ideal and rigid (no defects and void of phonon excitations) scenario where our particles (spinless bosons for now) will be acting, displaying all of their talents: kinetic energy and binary interactions. Our optical lattice is formed by a pair of laser beams that create an orthogonal standing wave ($\hat{V}_{\text{ext}}$) with orthogonal polarizations[29]. Because the optical lattice only allows for each one of the discrete holes to be filled from 0 to 2 particles, the interaction will be modeled by a contact potential (see Eq.A.8).

On a more serious note, our Hamiltonian in first quantization represented by single and two body operators is given by Eq. 2.1:

$$\hat{H} = \sum_{i=1}^{N} \hat{T}_i + \hat{V}_{ext} + \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} \hat{U}_{ij}. \tag{2.1}$$

Where $\hat{T}_i$, $\hat{V}_{ext}$ and $\hat{U}_{ij}$ are the kinetic energy of each particle, the periodic potential (lattice) and the interaction between the i-th particle and the j-th particle respectively. Furthermore, accounting for repeated terms inside the sum of the interacting terms such as U_{ij} and U_{ji} we've multiplied it by 0.5.

First quantization is not the most suitable formalism to analyze many-body systems. At first glance it provides little insight, and handling both one- and two-body operators can be cumbersome. For this reason, we turn to the more convenient second quantization framework[30].

In this representation, the Hamiltonian reads as in Eq. 2.2:

$$\hat{H} = \int dx, \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \nabla^2 + V_{ext}(x) \right) \hat{\Psi}(x) + \frac{g}{2} \int dx, dx', \hat{\Psi}^\dagger(x) \hat{\Psi}^\dagger(x') U(x, x') \hat{\Psi}(x') \hat{\Psi}(x). \tag{2.2}$$

For specifics the reader should go to Appendix A. After expressing the external potential as a discrete lattice and combining the single-particle term, see Eq. A.6, with the interaction term above, we arrive at the spinless Bose–Hubbard Hamiltonian (Eq. 2.3):

$$\hat{H} = -t \sum_{\langle i,j \rangle} \hat{b}_i^\dagger \hat{b}_j - \mu \sum_i \hat{n}_i + \frac{U}{2} \sum_i \hat{n}_i \left(\hat{n}_i - 1 \right). \tag{2.3}$$

2.2 Mott insulator - Superfluid Transition

In the Bose-Hubbard model at absolute zero, bosons confined in an optical lattice can realize two distinct phases depending on the relative strength of the competing energy scales. When the kinetic energy, encoded in the hopping amplitude t, dominates over the interactions, particles delocalize across the lattice and the system enters the *superfluid phase*[31]. Conversely, when the on-site interaction energy U outweighs tunneling, the particles become localized, number fluctuations are suppressed, and in the commensurate case the system forms a *Mott insulator*[32] with a fixed number of particles per site.

A central aspect of this transition is the competition between delocalization and localization, which is tuned by the ratio t/U. At big values of t/U, phase coherence extends throughout the system, while at small t/U number fluctuations vanish and insulating behavior sets in. The precise location of the critical point depends on dimensionality and filling, and can be characterized through observables such as the compressibility, the condensate fraction, and correlation functions that capture long-range order.

On the left-hand side of Eq. 2.4, one finds the description of a macroscopic wave function: the many-body problem effectively reduces to identical single-particle wave functions. This corresponds to the *superfluid regime*, where the Hamiltonian is dominated by kinetic energy ($t \gg U$) and quantum correlations play a crucial role. In contrast, on the right-hand side, when interactions prevail ($U \gg t$), tunneling processes are strongly suppressed and the interaction term becomes the only energetically relevant contribution. The system then enters a state with fixed particle numbers per site, the signature of the *Mott insulator*.

$$|\psi_{\mathrm{SF}}\rangle \propto \left(\sum_{i=1}^{L} \hat{b}_i^\dagger\right)^N |\phi\rangle \qquad\qquad |\psi_{\mathrm{MI}}\rangle \propto \prod_{i=1}^{L} \left(\hat{b}_i^\dagger\right)^N |\phi\rangle \tag{2.4}$$

The phase diagram in Fig. 2.1 illustrates the mean-field theory description of the Bose-Hubbard model. In this picture, the order parameter vanishes in the *Mott insulating* regime, while it acquires a finite value in the *superfluid* phase. The main focus of this thesis is the regime dominated by the

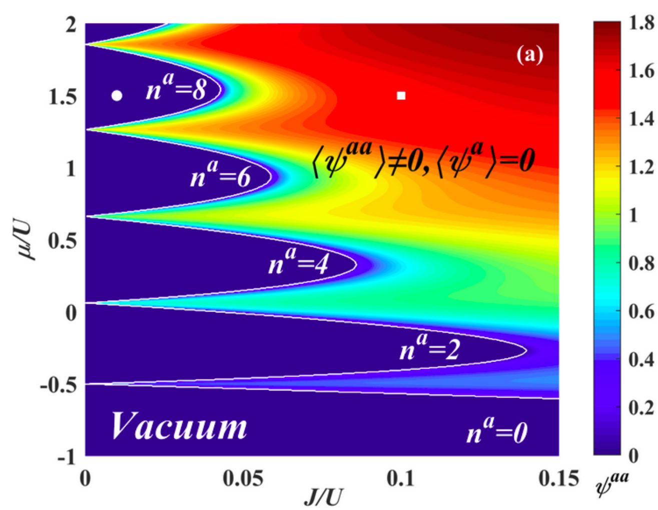


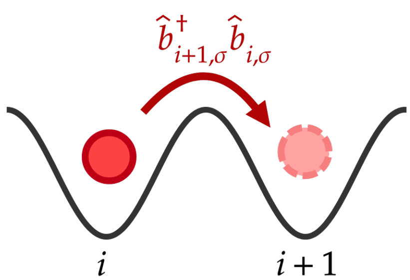
Figure 2.1: Ground-state phase diagram for the symmetric Bose-Hubbard model, adapted from[33]. The hopping parameter is denoted by $J = t$.

kinetic term, namely the *superfluid* phase, although the *Mott insulator* is also briefly examined in Chapter 4.

2.2.1 Introduction of Spin

An extra degree of freedom is what actually makes things interesting, the present work's system is the spin-1 BH Hamiltonian, the new spinorial degree of freedom brings up new rich physics and intriguing phenomena that's not present in the standard spinless or spin-$\frac{1}{2}$ cases. We'll have to consider how this new degree of freedom changes the interaction, as now the interacting potential in Eq. A.8 will also have to describe spinorial interactions[34].

To be able to build the spin-1 Hamiltonian we have to go over how the particles in the system manage to go from one site to another in the deep 1D lattice, and due to the fact that particles are confined to the lowest energy state at each location, mobility is only possible by tunneling between neighbor sites, as described by the hopping Hamiltonian in Eq. 2.5 as well as in figure 2.2.



$$\hat{H}_{\text{hop}} = -t\sum_{i,\sigma}\left(\hat{b}^\dagger_{i,\sigma}\hat{b}_{i+1,\sigma} + \hat{b}^\dagger_{i+1,\sigma}\hat{b}_{i,\sigma}\right). \tag{2.5}$$

Figure 2.2: Tunneling of a particle with spin σ from site i to its neighbor $i+1$.

Moreover, because of the bosonic symmetry, collisions can only take place in channels that are symmetric under particle exchange, corresponding to total spin $S = 0$ or $S = 2$. The antisymmetric $S = 1$ channel is therefore excluded. The coupled-basis states for the allowed and excluded channels are listed in Table A.1.

By replacing the projectors in Eq. A.11 in A.12 we get as a result Eq. 2.6, an expression for the interaction which is fairly easy to interpret, although quite different than the one we had in the spinless case (Eq. 2.1),

$$\hat{V}_{\text{int}} = \sum_i \Bigg[\frac{g_0 + 2g_2}{6} \hat{b}_0^\dagger \hat{b}_0^\dagger \hat{b}_0 \hat{b}_0 + \frac{2g_0 + g_2}{3} \hat{b}_1^\dagger \hat{b}_{-1}^\dagger \hat{b}_1 \hat{b}_{-1} + \frac{g_2 - g_0}{3} \left(\hat{b}_1^\dagger \hat{b}_{-1}^\dagger \hat{b}_0 \hat{b}_0 + \hat{b}_0^\dagger \hat{b}_0^\dagger \hat{b}_1 \hat{b}_{-1} \right) + \frac{g_2}{2} \left(\hat{b}_1^\dagger \hat{b}_1^\dagger \hat{b}_1 \hat{b}_1 + \hat{b}_{-1}^\dagger \hat{b}_{-1}^\dagger \hat{b}_{-1} \hat{b}_{-1} + 2\hat{b}_0^\dagger \hat{b}_{-1}^\dagger \hat{b}_0 \hat{b}_{-1} + 2\hat{b}_1^\dagger \hat{b}_0^\dagger \hat{b}_1 \hat{b}_0 \right) \Bigg]_i . \tag{2.6}$$

Two types of collisions can be identified in the system just by reading the previous equation, as expressed in figure 2.3, also their descriptions can be summed up to the following two processes:

I. **Spin-preserving collision:** Two particles (Fig. 2.3 I left) with an initial spin projection equal to σ_1, σ_2 are interacting (lines) and result in two particles (Fig. 2.3 I right) with the same spin projection.

II. **Spin-changing collision:** Two particles (Fig. 2.3 II left) with initial spin projections σ_1 and σ_2 interact (black shaded circle) and result in particles with spin σ_3 and σ_4, which are different than the initial ones (Fig. 2.3 II right).

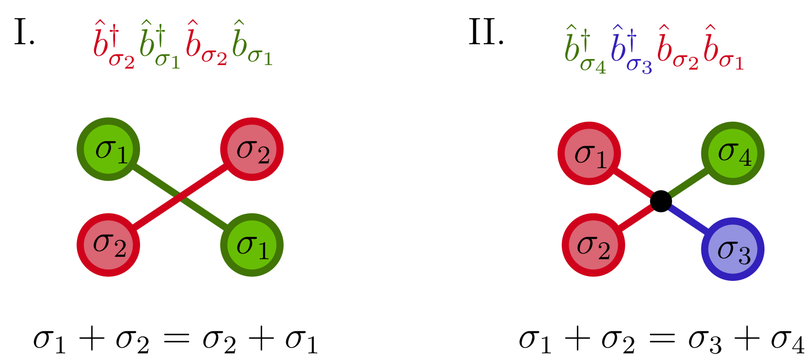


Figure 2.3: Interactions in the spin-1 Bose-Hubbard model, with $\sigma_i \in \{-1, 0, 1\}$. **(I)** Spin-preserving collision: two atoms with initial spin states σ_1 and σ_2 (shown in green and red) exchange their spins, yielding the final configuration (σ_2, σ_1). Total spin is conserved: $\sigma_1 + \sigma_2 = \sigma_2 + \sigma_1$. **(II)** Spin-changing collision: two atoms initially in spin states σ_1 and σ_2 scatter into a different pair σ_3 and σ_4, while also conserving total spin: $\sigma_1 + \sigma_2 = \sigma_3 + \sigma_4$.

Lastly, there's one final term we add to our Hamiltonian, a coupling to a linear Zeeman magnetic field[35], to lift the system's energy degeneracy between the different spin magnetic projections. The expression in second quantization is defined by Eq. 2.7,

$$\hat{H}_Z = B \sum_{i,\sigma} \sigma \hat{n}_{i,\sigma}, \tag{2.7}$$

where B is the strength of the linear Zeeman field. Thus we can sum all four terms that describe the dynamics of our system: The hopping term (Eq. 2.5), chemical potential, interaction (Eq. 2.6) and linear Zeeman field (Eq. 2.7) within with the second quantized representation of the spin-1 Bose-Hubbard model, as seen in Eq. 2.8,

$$\begin{aligned}\hat{H}_{\mathrm{BH}} &= \hat{H}_{\mathrm{hop}} + \hat{H}_{\mathrm{chem}} + \hat{H}_{\mathrm{Z}} + \hat{H}_{\mathrm{int}} \\ &= -t\sum_{i,\sigma}\left(\hat{b}^{\dagger}_{i,\sigma}\hat{b}_{i+1,\sigma} + \hat{b}^{\dagger}_{i+1,\sigma}\hat{b}_{i,\sigma}\right) - \mu\sum_{i,\sigma}\hat{n}_{i,\sigma} + B\sum_{i,\sigma}\sigma\hat{n}_{i,\sigma} \\ &\quad + \sum_{i}\left[\frac{g_0+2g_2}{6}\hat{b}^{\dagger}_0\hat{b}^{\dagger}_0\hat{b}_0\hat{b}_0 + \frac{2g_0+g_2}{3}\hat{b}^{\dagger}_1\hat{b}^{\dagger}_{-1}\hat{b}_1\hat{b}_{-1} + \frac{g_2-g_0}{3}\left(\hat{b}^{\dagger}_1\hat{b}^{\dagger}_{-1}\hat{b}_0\hat{b}_0 + \hat{b}^{\dagger}_0\hat{b}^{\dagger}_0\hat{b}_1\hat{b}_{-1}\right)\right. \\ &\quad \left. + \frac{g_2}{2}\left(\hat{b}^{\dagger}_1\hat{b}^{\dagger}_1\hat{b}_1\hat{b}_1 + \hat{b}^{\dagger}_{-1}\hat{b}^{\dagger}_{-1}\hat{b}_{-1}\hat{b}_{-1} + 2\hat{b}^{\dagger}_0\hat{b}^{\dagger}_{-1}\hat{b}_0\hat{b}_{-1} + 2\hat{b}^{\dagger}_1\hat{b}^{\dagger}_0\hat{b}_1\hat{b}_0\right)\right]_i .\end{aligned} \tag{2.8}$$

Much like the spinless Bose-Hubbard model, the spin-1 extension supports both superfluid and Mott insulating phases, controlled primarily by the competition between the tunneling amplitude t and the on-site (g_0, g_2) interactions. The hopping term remains independent of the spin degree of freedom, even in the presence of linear or quadratic Zeeman fields, since these fields act along the z direction and therefore do not modify the amplitude with which particles tunnel between neighboring sites. The essential difference with respect to the spinless case arises in the interaction sector, particles now interact not only through their total on-site occupation, but also through their internal spin. This yields the spin-dependent interaction term of Eq. 2.6, which couples different spin channels and allows for nontrivial spin mixing and conversion processes absent in the spinless model. As a consequence, the spin-1 Bose-Hubbard model exhibits a much richer interplay between charge and spin degrees of freedom while retaining the familiar superfluid-Mott phenomenology at the level of its phase structure.

To have a clearer picture on what's the linear Zeeman field doing to our system it's useful to transform only the single-particle Hamiltonian in the momentum space, due to the translational symmetry[29] we can use Eq. 2.9,

$$\hat{b}_{i,\sigma} = \frac{1}{\sqrt{L}}\sum_{k} e^{ikl}\hat{b}_{k,\sigma}, \tag{2.9}$$

to take a look in the dispersion relation, $\varepsilon_\sigma(k)$, as seen in Eq. 2.10,

$$\hat{H}_0 = \sum_{k,\sigma} \underbrace{(\mu - \sigma B + 2t\cos k)}_{\varepsilon_\sigma(k)} \hat{n}_{k,\sigma}, \tag{2.10}$$

the dispersion relation is plotted in Figure 2.4. Panel (A) illustrates that, in the absence of an external magnetic field, the three spin projections $\sigma = -1, 0, 1$ are energetically equivalent, and the system is therefore triply degenerate. In Panels (B)-(D), the application of a magnetic linear Zeemann field B shifts the energies of the spin components according to their magnetic projection. When $B > 0$, the $\sigma = 1$ branch is shifted downward in energy, making it the most energetically favorable, while $\sigma = 0$ and $\sigma = -1$ are pushed higher, Whereas, for $B < 0$, the $\sigma = -1$ branch is favored over $\sigma = 0, 1$.

Having introduced the theoretical framework of the Bose–Hubbard model in Eq. 2.8, we now turn to the problem of solving it. In this work, we choose a numerical approach based on Tensor Networks methods, the following chapter introduces the mathematical formalism behind these methods, complemented by a diagrammatic representation using Penrose notation to provide intuitive insight into their structure and construction.

There are three main aims in this thesis. First, to study the physics of a one-dimensional system of interacting spin-1 lattice bosons subject to an external Zeeman magnetic field, described by the aforementioned BH model in the present chapter. Second, to implement a numerical method based on Tensor Networks algorithms that exploits the translational symmetry of the periodic lattice (Chapter 3). And third, to characterize the superfluid phase through the analysis of correlations and local observables, therefore examining the magnetic properties of the system in the presence of a linear magnetic field, with results reported in Chapter 4, including the identification of Bose–Hubbard states in the Mott insulating regime such as the Ising and dimer phases.

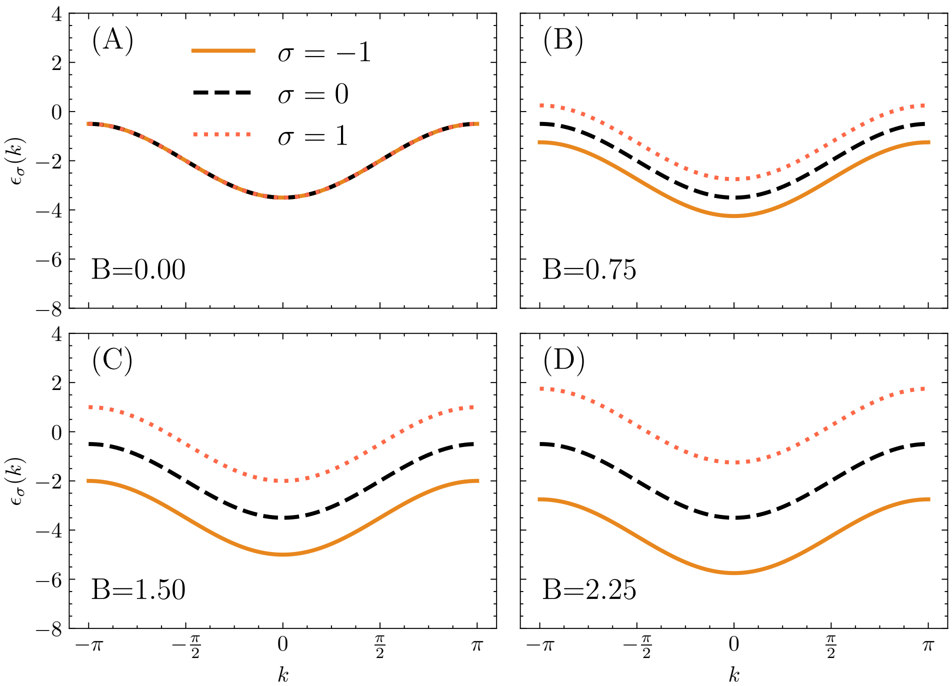


Figure 2.4: Dispersion relation $\varepsilon_\sigma(k)$ for different values of the magnetic field B with fixed values $t = 0.75$, $\mu = 2$.

Chapter 3

Tensor Networks

Quantum many body physics is the name of the game, and understanding it is our challenge, plenty of phenomena in condensed matter physics such as exotic phases of quantum matter has many mysteries unraveling it, mostly because of the fact that it has a tough and complicated analytical approach due to the mere size the systems scale to, even a model as simple as the Hubbard which is exactly solvable only in 1D via the Bethe ansatz [36, 37] requires of numerical methods [38, 39] to determine with rich insights their properties for general or more complicated systems.

Making a long story short, tensor networks (TN) [40] methods relies on the fact that the highly correlated wave function that describes a given system may be broken down in tensors, in the same way DNA describes a living being, tensors describe the wave function.

There are a diversity of numerical methods to analyze strongly correlated systems, why choosing TN out of all the other? The answer, although not trivial, is based on the fact that every existing method has its limitations, for instance, Exact Diagonalization [41] of the Hamiltonian is limited to small-scale systems, leaving bulk properties close to the thermodynamic limit almost out of its reach, Quantum Mote Carlo [42] is handicapped at the door of systems such as Fermionic and frustrated quantum spin systems, due to the sign problem, and Mean Field Theory [43, 44] has issues regarding precise calculations of quantum correlations at low dimensions.

Although TN isn't perfect either, it suits perfectly with the type of system we are going to tackle (specifically gapped 1D Hamiltonians), as the limitations are quite different from the rest, the amount and structure of the entanglement in quantum many-body states is a feature, instead of an inconvenience that has to be worked around.

Cutting to the chase, TN describe quantum many-body states as networks of interconnected tensors, a structure that makes correlations and entanglement explicit. To work with these objects efficiently, we introduce Penrose graphical tensor notation (see Fig. 3.2), which provides a clear and intuitive diagrammatic language for representing tensors and their contractions. This notation allows us to manipulate and visualize tensor-network states in a way that reflects their underlying structure directly, and it will serve as the basis for introducing the Matrix Product State (MPS) ansatz (see

Section 3.2).

The principal reason to use TN to describe a quantum many-body state lies in the rapid growth of the dimension of the Hilbert space, as the number of sites (or degrees of freedom) increase. For instance, given d degrees of freedom and N particles, the dimensions of the Hilbert space describing such system would be d^N which grows rapidly as either the degrees of freedom or the number of particles increase, for example, in a system of spin-$\frac{1}{2}$ fermions, it would only take about 57 particles to reach a Hilbert space bigger than the amount of seconds in the age of the universe (about 10^{17}), considering that a mol of substance has around 10^{23}, which leaves one wondering what to do with that many quantum states. Fortunately, not all quantum states in a Hilbert space of a many-body system are actually important, some are way more relevant than others, taking into account that our Hamiltonian has local interactions between particles and lattices sites, there's an important consequence to that, and is that the entanglement entropy of the ground/low energy states of that kind of Hamiltonians tends to scale as the size of the boundary of the region, and not as the volume, this is known as the *area-law* [40] (See Fig. 3.1 (I)).

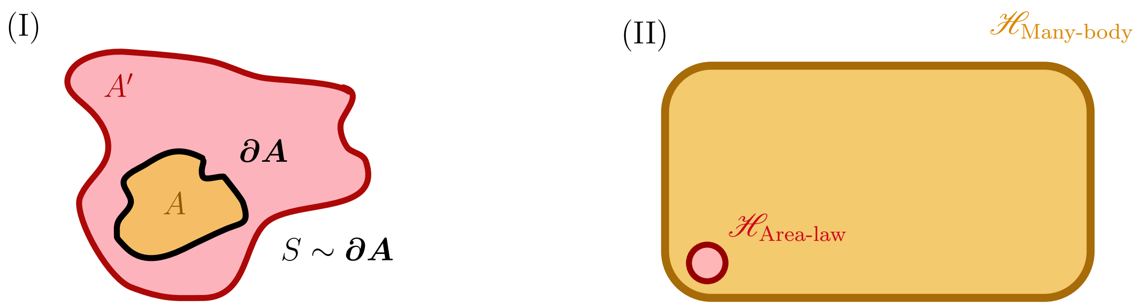


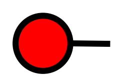
Figure 3.1: (A) Illustration of the area law for entanglement entropy. For a subsystem A with complement A', the entanglement entropy S typically scales with the size of the boundary ∂A, rather than with the volume of A. (B) Representation of the Hilbert space of many-body states where the physically relevant states occupy only an exponentially small corner of the full Hilbert space.

The reason why this is such an important property is that if we pick a quantum state out of our huge Hilbert space at random, it will almost certainly exhibit entanglement entropy between subregions that scales with the volume of the subsystem [45]. In contrast, physically relevant low-energy states of local Hamiltonians the entanglement entropy scales with the size of the boundary between regions rather than the volume of the bulk [46, 47]. In one dimension, the area law is rigorously proven for ground states of gapped Hamiltonians [46], while in higher dimensions it is widely believed to hold, up to possible logarithmic corrections at criticality. This dramatic difference reduces the set of physically relevant states to a tiny corner of Hilbert space, exponentially smaller than the full space, as depicted in panel (II) of Fig. 3.1.

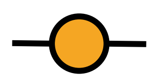
3.1 Tensor Network diagrams

For the sake of introducing the key concepts which TN revolves around, a tensor is a multidimensional array of complex numbers, whose rank corresponds to the number of indices. Throughout the calculations that have to be made either to calculate expected values or orthonormalize an initial state, a graphical notation known as Penrose Diagrams is introduced, where the tensors are treated as geometrical figures, and the lines are the number of indices it has, as shown in Fig 3.2.

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \longleftrightarrow \boldsymbol{v_i}$$



$$M = \begin{pmatrix} m_1 & m_2 & m_3 \\ m_4 & m_5 & m_6 \\ m_7 & m_8 & m_9 \end{pmatrix} \longleftrightarrow \boldsymbol{M_{ij}}$$



$$\overleftrightarrow{T} = \begin{pmatrix} t_{11} & t_{12} & t_{13} \\ t_{14} & t_{15} & t_{16} \\ t_{17} & t_{18} & t_{19} \end{pmatrix} \begin{pmatrix} t_{21} & t_{22} & t_{23} \\ & & t_{26} \\ & & t_{29} \end{pmatrix} \begin{pmatrix} t_{31} & t_{32} & t_{33} \\ & & t_{36} \\ & & t_{39} \end{pmatrix} \longleftrightarrow \boldsymbol{T_{ijk}}$$

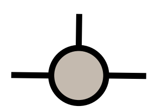


Figure 3.2: From top to bottom, the graphic representation in Penrose's notation of a vector, matrix and 3-tensor.

Among the important procedures/algebraic manipulations we have contractions (see Figure 3.3), index regrouping, singular value decomposition (SVD) among others. In the case of the last one, a key figure is selected to represent if a tensor is orthogonalized by either rows or columns, as seen in figure 3.4.

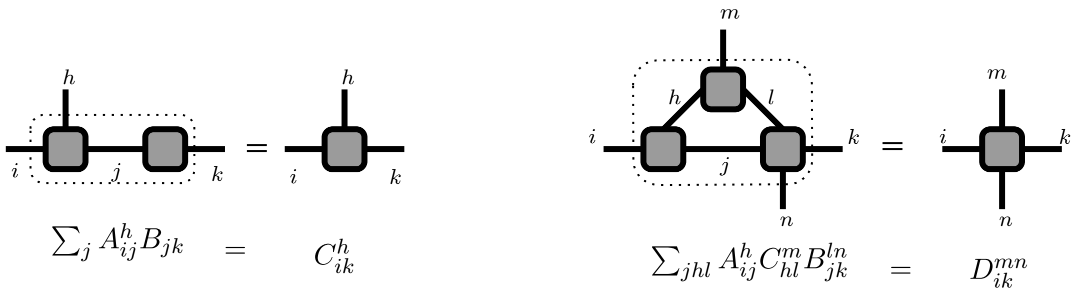


Figure 3.3: Contraction of indexes in Penrose graphical notation.

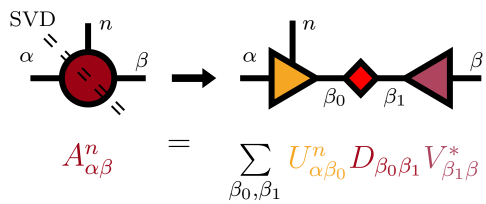


Figure 3.4: Graphical representation of the singular value decomposition of the 3-tensor A, where U and V are orthogonal matrices, the first is composed by columns which are left singular vectors of A, while V's rows are right singular vectors of A.

3.2 Matrix Product States

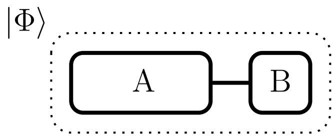
Now letting the graphical representation out of the way for a little while, and remembering that any state can be represented by the superposition it's local basis elements and some coefficients which quantifies the probability amplitude $c_i = \langle\Phi|\phi_i\rangle$ of finding that state $|\Phi\rangle$ in that specific basis element $|\phi_i\rangle$, as

$$|\Phi\rangle = \sum_i c_i |\phi_i\rangle ,$$

let's consider a multipartite 1D quantum system which can be represented as a chain of L sites, each with internal dimension d that's described by a state $|\Phi\rangle$ [48]. To construct the total Hilbert space $\mathcal{H}$ such that $|\Phi\rangle \in \mathcal{H}$ then the Hilbert space $\mathcal{H}^i$ of each site i must be taken into account as well, as

$$\mathcal{H} = \bigotimes_{i=1}^{L} \mathcal{H}^i$$

taking into account that the dimension of each $\mathcal{H}^i$ is d, then the dimension of the total Hilbert space must be d^L. Taking the most general scenario with the most entanglement, we'll treat the state $|\Phi\rangle$ as irreducible to the direct product of states belonging to different sites. However, we can in fact express a bipartite system whose Hilbert space can be expressed as the tensor product of the two subspaces $\mathcal{H} = \mathcal{H}^A \otimes \mathcal{H}^B$, where $\{|A_i\rangle\} \in \mathcal{H}^A$ and $\{|B_i\rangle\} \in \mathcal{H}^B$ form a complete set of orthonormal basis vectors and therefore we can express $|\Phi\rangle$ as an entangled state:



$$|\Phi\rangle = \sum_{i}^{N_A} \sum_{j}^{N_B} C_{ij} |A_i\rangle \otimes |B_i\rangle , \tag{3.1}$$

given the structure of our Hilbert space, we can express any state using the Schmidt decomposition [48]

$$|\Phi\rangle = \sum_{\alpha_1}^{\chi} \lambda_{\alpha_1} |\alpha_1\rangle |\omega_{\alpha_1}\rangle , \tag{3.2}$$

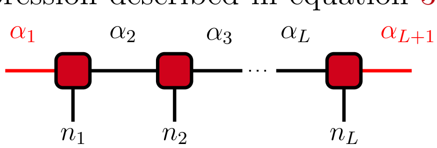
where $N_A > N_B$ and $\chi \equiv N_A$, and λ_i are known as Schmidt coefficients, are non-negative and follow the following relation $\sum_i \lambda_i^2 = 1$. A single site $|\omega_{\alpha_i}\rangle$ is left isolated, and can be transformed into a known local basis $\{n_i\}$ (i.e Fock basis, Magnetization, or other) as expressed in the left hand side of equation 3.3. The right hand side expression describes essentially that $|\alpha_i\rangle$ can be decomposed as well, leaving another isolated site and a new $|\alpha_{i-1}\rangle$.

$$|\omega_{\alpha_i}\rangle = \sum_{n_i}^{d} \Gamma^{n_i}_{\alpha_i \alpha_{i+1}} |n_i\rangle , \qquad |\alpha_{i+1}\rangle = \sum_{\alpha_i}^{\chi} \lambda_{\alpha_i} |\alpha_i\rangle |\omega_{\alpha_i}\rangle . \tag{3.3}$$

By joining the two expressions in equation 3.3 we get a better representation of the bipartite state in terms of the local basis, as seen in equation 3.4.

$$\begin{aligned} |\alpha_{i+1}\rangle &= \sum_{\alpha_i}^{\chi} \sum_{n_i}^{d} \lambda_{\alpha_i} \Gamma^{n_i}_{\alpha_i \alpha_{i+1}} |\alpha_i\rangle |n_i\rangle , \\ &= \sum_{\alpha_i}^{\chi} \sum_{n_i}^{d} A^{n_i}_{\alpha_i \alpha_{i+1}} |\alpha_i\rangle |n_i\rangle , \end{aligned} \tag{3.4}$$

where $A^{n_i}_{\alpha_i \alpha_{i+1}} \equiv \lambda_{\alpha_i} \Gamma^{n_i}_{\alpha_i \alpha_{i+1}}$. By replacing this recursion relation in equation 3.13 we'll reach the expression described in equation 3.5, to have a clearer view on the step by step see Figure 3.5.



$$|\Phi\rangle = \sum_{\alpha_1 \cdots \alpha_L}^{\chi} \sum_{n_1 \cdots n_L}^{d} A^{n_1}_{\alpha_1 \alpha_2} A^{n_2}_{\alpha_2 \alpha_3} \dots A^{n_L}_{\alpha_L \alpha_{L+1}} |n_1 \cdots n_L\rangle . \tag{3.5}$$

There are a couple of nuances with α_1 and α_L, as they are the first and the last sites of the chain, their A's are originally rank-2 tensors, but a dummy index $\alpha_1 = 1$ and $\alpha_{L+1} = 1$ are added to keep the same structure throughout the state coefficient.

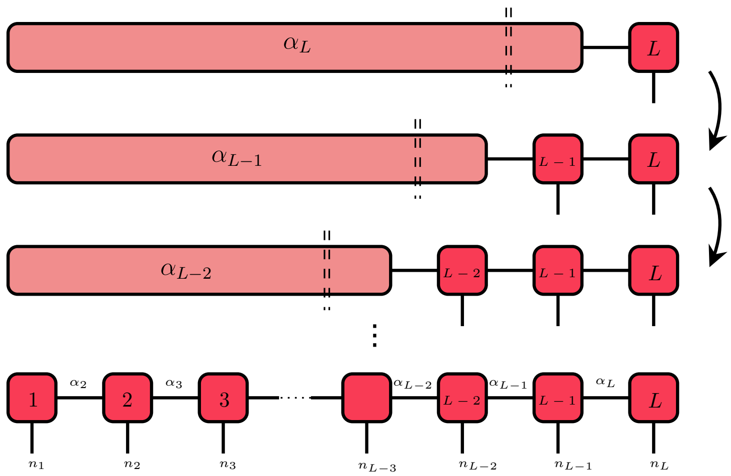


Figure 3.5: Schmidt decomposition of state in equation 3.13, replacing equation 3.4 after each iteration. SVD is represented as double dashed lines.

3.3 Canonical forms of a MPS

The expression we've reached in equation 3.5 isn't the most convenient one for our purposes, it can be further simplified to have better performance while the computation of expectation values. Although the representation is really chosen at the convenience of the programmer due to MPS exhibiting a gauge freedom that allows for infinitely many equivalent representations. Specifically, for any bond index i, one can introduce an arbitrary unitary matrix U and its inverse $U^\dagger$, transforming the MPS tensors without altering the physical state, as seen in Figure 3.6. In fact, this gauge freedom [48] is precisely what enables us to impose a *canonical form* which will optimize computational performance. The three most common canonical forms are presented in Figure 3.7, nevertheless, the *mixed canonical form* presented in equation 3.6 is the one we'll be using, as it leaves the k-th site matrix untouched, granting us an orthogonality center.

$$|\Phi\rangle = \sum_{\alpha_k \alpha_{k+1}}^{\chi} \sum_{n_k}^{d} A^{n_k}_{\alpha_k \alpha_{k+1}} |\alpha_k\rangle |n_k\rangle |\omega_{\alpha_{k+1}}\rangle . \tag{3.6}$$

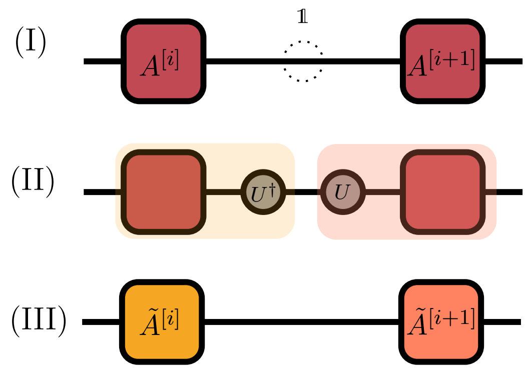


Figure 3.6: Gauge freedom in a MPS. (I) The tensors $A^{[i]}$ and $A^{[i+1]}$ can be transformed (II) by inserting an arbitrary invertible matrix U and its inverse $U^\dagger$ between the bond indices. (III) This transformation results in new tensors $\tilde{A}[i] = A^{[i]}U$ and $\tilde{A}^{[i+1]} = U^{-1}A^{[i+1]}$, leaving the overall state unchanged. This gauge flexibility is fundamental for imposing canonical forms and optimizing numerical algorithms.

The explicit expression for $|\alpha_k\rangle$ and $\left|\omega_{\alpha_{k+1}}\right\rangle$ in terms of column-orthogonal and row-orthogonal unitary matrices $U^\dagger U = \mathbb{1}$ and $VV^\dagger = \mathbb{1}$ can be found in equation 3.7.

$$|\alpha_k\rangle = \sum_{\alpha_1\cdots\alpha_{k-1}} \prod_{s=1}^{k-1} U^{n_s}_{\alpha_s\alpha_{s+1}} |n_s\rangle, \qquad \left|\omega_{\alpha_{k+1}}\right\rangle = \sum_{\alpha_{k+1}\cdots\alpha_L} \prod_{s=k+1}^{L} V^{n_s *}_{\alpha_s\alpha_{s+1}} |n_s\rangle. \tag{3.7}$$

It's crucial to keep in mind that they form two sets of orthonormal basis such that $\langle\alpha_k|\alpha'_k\rangle = \left\langle\omega_{\alpha_k}\middle|\omega_{\alpha'_k}\right\rangle = \delta_{\alpha_k\alpha'_k}$, thus leaving the state $|\Phi\rangle$ orthonormalized! The exact procedure to get the mixed

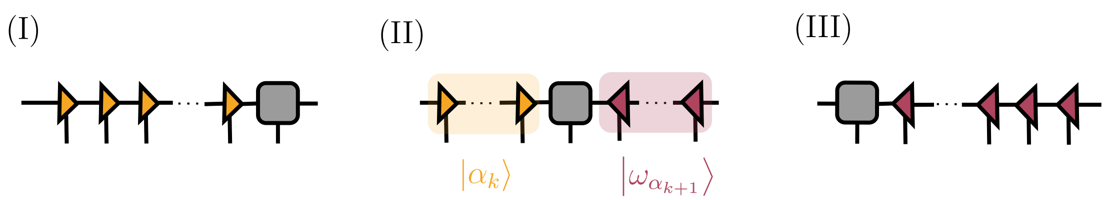


Figure 3.7: Canonical representations of a MPS. (I) left, (II) Mixed, (III) Right canonical forms.

MPS canonical representation is shown in Figure 3.8, by doing the singular value decomposition of each tensor $A^{n_s}_{\alpha_s\alpha_{s+1}}$ just as in Figure 3.4 and then contracting D as well as V^* to the neighbor $A^{n_{s+1}}_{\alpha_{s+1}\alpha_{s+2}}$, becoming $\tilde{A}^{n_{s+1}}_{\alpha_{s+1}\alpha_{s+2}}$, which has to be decomposed in its singular values and repeat the aforementioned process.

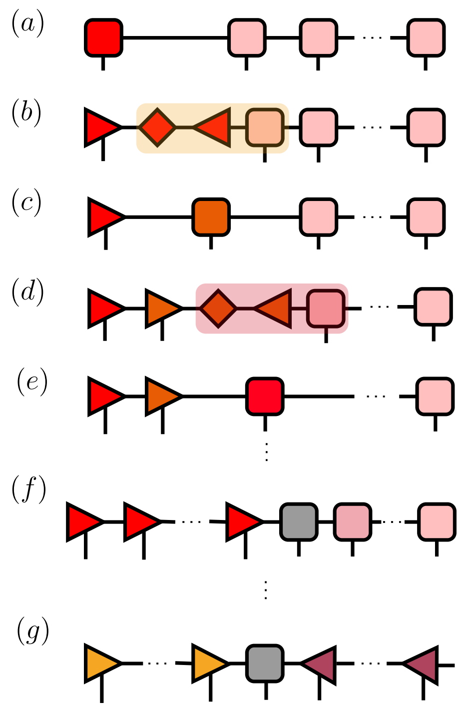


Figure 3.8: MPS canonical mixed representation procedure. (a) Starting from the original MPS a (b) singular value decomposition of $A^{n_1}_{\alpha_2}$ is made, leaving three matrices $U_{\alpha'_2}$, $D_{\alpha'_2\alpha''_2} = D_{\alpha'_2}\delta_{\alpha'_2\alpha''_2}$ and $V^*_{\alpha''_2\alpha_2}$. (c) Contracting D, V^* and $A^{[2]}$ into $\tilde{A}^{[2]}$. (d) Taking the SVD of the newly contracted tensor, to leave a new U matrix behind and (e) contract D, V and $A^{[3]}$. (f) Iterate the aforementioned procedure until the k-th site (shaded gray). (g) At last, the process is repeated for the left side of the MPS, from L until $k+1$, where instead of leaving row-orthogonal matrices U, column-orthogonal matrices V^* compose that wing.

3.4 Expectation values

Now that we have an orthonormalized MPS let's get to the thick of it, doing measurements and calculating expectation values, as with the right observables we'll get to know our system's behavior.

3.4.1 Local operators

First things first, to calculate a general single site acting operator $\left\langle \hat{\mathcal{O}}_k \right\rangle$ on the k-th site, the same one where our orthogonality center is, as showcased in Figure 3.9. The mathematical expression can be found in equation 3.8, where due to the state's orthonormality, we end up only needing a single calculation rather than one per site.

$$
\begin{aligned}
\langle\Phi|\,\hat{\mathcal{O}}_k\,|\Phi\rangle &= \left(\sum_{\alpha'_k\alpha'_{k+1}}^{\chi}\sum_{n'_k}^{d} A^{n'_k *}_{\alpha'_k\alpha'_{k+1}}\,\langle\alpha'_k|\,\langle n'_k|\,\left\langle\omega_{\alpha'_{k+1}}\right|\right)\hat{\mathcal{O}}_k\left(\sum_{\alpha_k\alpha_{k+1}}^{\chi}\sum_{n_k}^{d} A^{n_k}_{\alpha_k\alpha_{k+1}}\,|\alpha_k\rangle\,|n_k\rangle\,\left|\omega_{\alpha_{k+1}}\right\rangle\right)\\
&= \sum_{\substack{\alpha'_k\alpha'_{k+1}\\ \alpha_k\alpha_{k+1}}}^{\chi}\sum_{n'_k n_k}^{d} A^{n'_k *}_{\alpha'_k\alpha'_{k+1}}\underbrace{\langle n'_k|\,\hat{\mathcal{O}}_k\,|n_k\rangle}_{\mathcal{O}_{n'_k n_k}} A^{n_k}_{\alpha_k\alpha_{k+1}}\underbrace{\langle\alpha'_k|\alpha_k\rangle}_{\delta^{\alpha'_k}_{\alpha_k}}\underbrace{\left\langle\omega_{\alpha_{k+1}}\middle|\omega_{\alpha'_{k+1}}\right\rangle}_{\delta^{\alpha'_{k+1}}_{\alpha_{k+1}}}\\
&= \sum_{\substack{\alpha'_k\alpha'_{k+1}\\ \alpha_k\alpha_{k+1}}}^{\chi}\underbrace{\sum_{n'_k n_k}^{d} A^{n'_k *}_{\alpha'_k\alpha'_{k+1}}\,\mathcal{O}_{n'_k n_k} A^{n_k}_{\alpha_k\alpha_{k+1}}}_{\mathbb{E}(\hat{\mathcal{O}}_k)}\,\delta^{\alpha'_k}_{\alpha_k}\delta^{\alpha'_{k+1}}_{\alpha_{k+1}}\\
&= \sum_{\substack{\alpha'_k\alpha'_{k+1}\\ \alpha_k\alpha_{k+1}}}^{\chi}\mathbb{E}(\hat{\mathcal{O}}_k)\delta^{\alpha'_k}_{\alpha_k}\delta^{\alpha'_{k+1}}_{\alpha_{k+1}} = \mathbf{Tr}\left(\mathbb{E}(\hat{\mathcal{O}}_k)\delta^{\alpha'_k}_{\alpha_k}\delta^{\alpha'_{k+1}}_{\alpha_{k+1}}\right)
\end{aligned}
\tag{3.8}
$$

A brilliant contraction appears, and it's most likely the most fundamental part of the whole MPS'

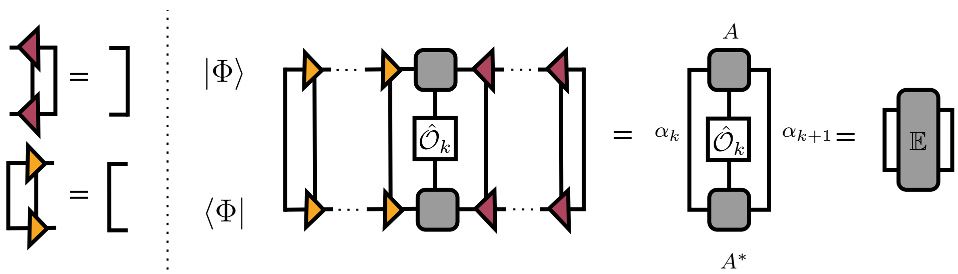


Figure 3.9: Graphical representation of the expectation value of a local operator in an MPS, leading to the definition of the transfer matrix $\mathbb{E}$.

development, the transfer matrix $\mathbb{E}(\hat{\mathcal{O}}_k)$, whose trace over α_k and α_{k+1} yields the expected value of $\hat{\mathcal{O}}_k$. Secondly, there are other cases of interest, as an example Figure 3.10 operators which don't act necessarily on the center of orthogonality k, but on a site $s > k$, the amount of contractions as well as the number of transfer matrices increase, but equation 3.9 shows what the final expression is, if interested on the whole operation, the reader may head to Appendix B.

$$
\langle\Phi|\,\hat{\mathcal{O}}_s\,|\Phi\rangle = \mathbf{Tr}\left[\mathbb{E}(\hat{\mathbb{I}}^k)\mathbb{E}(\hat{\mathbb{I}}^{k+1})\cdots\mathbb{E}(\hat{\mathcal{O}}_s)\right]. \tag{3.9}
$$

Notice that if $\hat{\mathcal{O}} = \hat{1}$ then what we'll be obtaining corresponds to the norm of the state $\langle\Phi|\Phi\rangle$.

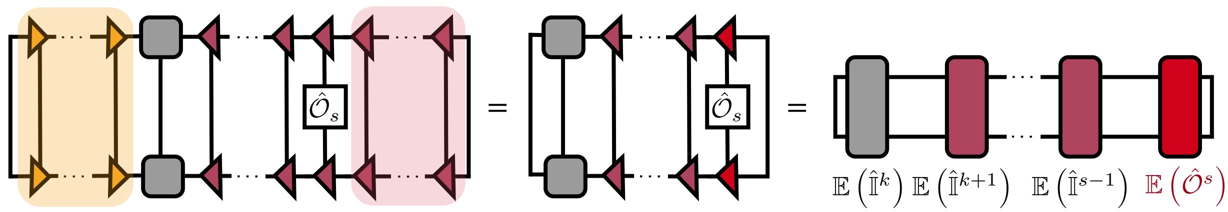


Figure 3.10: Graphical representation of the expected value of an operator $\hat{\mathcal{O}}$ in a site s away from the center of orthogonality of the MPS.

3.4.2 Non-local operators: Correlations

Let's move on to the calculation of the correlation $\left\langle \hat{\mathcal{O}}_1^k \hat{\mathcal{O}}_2^{k+\Delta} \right\rangle$, which has two different operators action on both k-th site and Δ sites to the right. It's crucial to calculate correlations, as understanding them is foundational to really comprehending our physical system, they capture the essence of interactions and rich phenomena. As it is often attributed to John A. Wheeler, though not verbatim in his published writings, "*physics is the study of correlations.*" [49].

Fortunately, the expression it takes in equation 3.10 is quite similar to the one we already found in equation 3.9, and the same can be said for their graphical representation, as seen in Figure 3.11, where two transfer matrices of each site operator, k and $k+\Delta$, must be computed and the product of identity-transfer matrices in between,

$$\left\langle \hat{\mathcal{O}}_1^k \hat{\mathcal{O}}_2^{k+\Delta} \right\rangle = \langle \Phi | \, \hat{\mathcal{O}}_1^k \hat{\mathcal{O}}_2^{k+\Delta} \, | \Phi \rangle = \mathbf{Tr} \left[\mathbb{E}(\hat{\mathcal{O}}_1^k) \left(\prod_{j=k+1}^{k+\Delta-1} \mathbb{E}(\hat{\mathbb{I}}^j) \right) \mathbb{E}(\hat{\mathcal{O}}_2^{k+\Delta}) \right]. \tag{3.10}$$

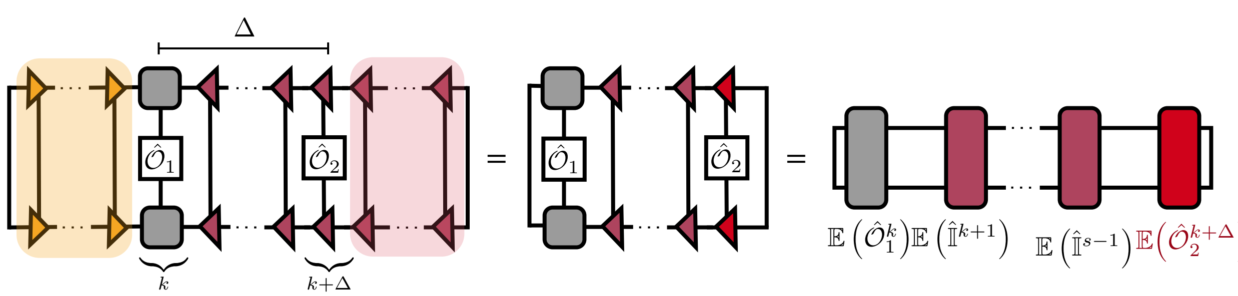


Figure 3.11: Graphical representation of a two-point correlation function in an MPS. The insertion of local operators $\hat{\mathcal{O}}_1$ and $\hat{\mathcal{O}}_2$ at sites k and $k+\Delta$ reduces to a product of transfer matrices, with Δ encoding the separation between operators.

3.5 Matrix Product Operators

Now, to the meat and bone of this work! This section presents the main contribution to our theoretical physics research group. Previous MPS approaches faced limitations related to system scalability, basis growth, and over-completeness, issues that are addressed in more detail in Sec. 3.6.

As one can represent a quantum state in terms of products of matrices, the same can be done with any given operator. In fact, we are particularly interested in expectation values of both non-local operators and single-site ones acting along the entire chain, most notably the system's Hamiltonian, which naturally leads to a Matrix Product Operator (MPO) representation [50]. Following the same philosophy of the already discussed MPS, it is desirable to express operators as a product of matrices, as shown in equation 3.11 for a general operator.

$$
\begin{aligned}
\hat{O} &= \sum_{\{n\},\{n'\}}^{d} C_{n_1 n_2 \cdots n_L}^{n'_1 n'_2 \cdots n'_L} \left| n_1 n_2 \cdots n_L \right\rangle \left\langle n'_1 n'_2 \cdots n'_L \right| \\
&= \sum_{\{n\}\{n'\}}^{d} \sum_{\{\beta\}}^{w} W_{\beta_1} W_{\beta_1 \beta_2}^{n_1 n'_1} W_{\beta_2 \beta_3}^{n_2 n'_2} \cdots W_{\beta_{L-1} \beta_L}^{n_{L-1} n'_{L-1}} W_{\beta_L \beta_{L+1}}^{n_L n'_L} W_{\beta_{L+1}} \left| n_1 \cdots n_L \right\rangle \left\langle n'_1 \cdots n'_L \right| \\
&= \sum_{\{n\}\{n'\}}^{d} W^{[0]} W^{[1]} W^{[2]} \cdots W^{[L-1]} W^{[L]} W^{[L+1]} \left| n_1 \cdots n_L \right\rangle \left\langle n'_1 \cdots n'_L \right| .
\end{aligned}
\tag{3.11}
$$

In equation 3.11 there's quite a lot to unpack, the element $W_{\beta_k \beta_{k+1}}^{[k]}$ is a $w \times w$ matrix and $\{|n_k\rangle\}$ are the local basis vectors. We can perform a matrix multiplication over all of the tensors W and we'll be left with a scalar. The graphic representation of equation 3.11 can be seen in Figure 3.12. The true wonders of the MPO representation lies in the fact that it allows us to represent *exactly* any local Hamiltonian with (relatively) small matrices of dimension $w \times w$, also that their structure is almost identical that of a MPS, where the two of them are represented by a list of tensors at each site and, therefore, computing expectation values becomes an easier task! Taking into account that we're working with a Hamiltonian which is composed by a sum of finite-range interactions we can construct the W matrix as entirely lower (or upper) triangular matrices, by imposing that the top-left and bottom-right components are identity operators $\hat{1}$. Let's start with a given operator which for futur purposes we take the system Hamiltonian, that's the sum of single

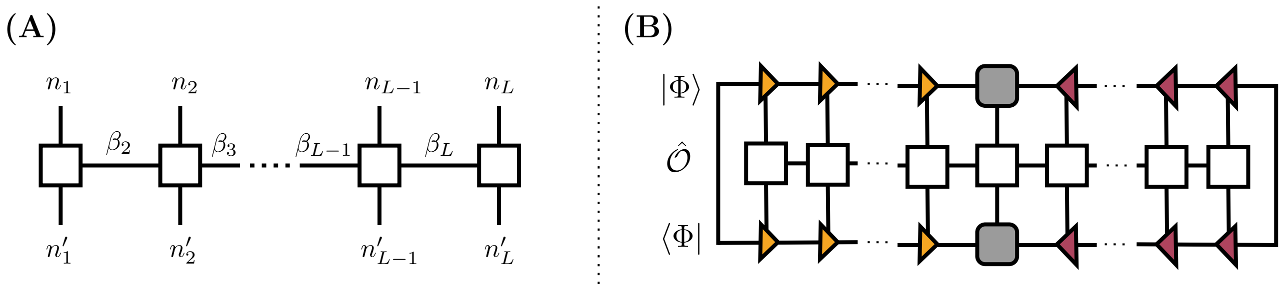


Figure 3.12: **(A)** Tensor network diagram of a MPO acting on a chain of L sites, with physical indices n_i and n'_i and bond dimensions β_i. **(B)** Expectation value of an MPO in a MPS, obtained by contracting the MPO tensors with the MPS bra and ket.

site local terms:

$$\hat{H} = \sum_i \hat{X}_i = W^{[1]}W^{[2]}\cdots W^{[L-1]}W^{[L]} \quad \longrightarrow \quad W^{[k]} = \begin{bmatrix} \hat{1} & \hat{0} \\ \hat{X}_k & \hat{1} \end{bmatrix},$$

for that specific Hamiltonian, the internal dimension of the MPO is $w = 2$, and each operator (including that top right 0) are $d \times d$ matrices. If we do the same procedure for a nearest-neighbor Hamiltonian that also have single site local terms we get:

$$\hat{H} = \sum_i \hat{X}_i\hat{Y}_{i+1} + \sum_j \hat{Z}_j = W^{[1]}W^{[2]}\cdots W^{[L-1]}W^{[L]} \quad \longrightarrow \quad W^{[k]} = \begin{bmatrix} \hat{1} & \hat{0} & \hat{0} \\ \hat{Y}_k & \hat{0} & \hat{0} \\ \hat{Z}_k & \hat{X}_k & \hat{1} \end{bmatrix},$$

notice that in both the first and the second Hamiltonian the terms which act upon a single site always remain in the bottom left. We can also express our Spin-1 BH Hamiltonian in the way seen in equation 3.12. If the reader is interested in how to make an MPO of their own they can head to

Appendix appB to have a clearer picture.

$$W^{[i]} = \begin{bmatrix} \hat{1} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}_{i,1} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}^{\dagger}_{i,1} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}_{i,0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}^{\dagger}_{i,0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}_{i,-1} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ -t\hat{b}^{\dagger}_{i,-1} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} & \hat{0} \\ \hat{H}_{\text{int}} & \hat{b}^{\dagger}_{i,1} & \hat{b}_{i,1} & \hat{b}^{\dagger}_{i,0} & \hat{b}_{i,0} & \hat{b}^{\dagger}_{i,-1} & \hat{b}_{i,-1} & \hat{1} \end{bmatrix}. \tag{3.12}$$

Also, the matrices at the beginning and at the end of the chain will have quite a different structure, they aren't square matrices, instead they are column and row vectors of dimension $1 \times w$ and $w \times 1$, so that the product between W's yield a scalar, see equation 3.13.

$$W^{[1]} = \begin{bmatrix} 0 & \cdots & \hat{0} & \hat{1} \end{bmatrix}, \qquad W^{[L]} = \begin{bmatrix} \hat{1} & \hat{0} & \cdots & \hat{0} \end{bmatrix}^T. \tag{3.13}$$

3.6 Numerical implementation

Up until now, it's been detailed how the different elements in the TN approach can be written down as; nevertheless, to be able to put that into practice it's necessary to find an appropriate representation in the computer as well. There are widely known libraries for tensor networks calculations such as DMRJulia [51], cuTensorNet [52] and iTensor [53]. Although existing libraries provide excellent implementations with a wide range of models and efficient tensor contractions, we have chosen a more hands-on approach, manually reshaping tensors and performing the contractions ourselves. This pedagogical choice allows us to better understand the inner workings of large scale libraries, while also continuing a library previously developed by A. Arguelles and K. Rodríguez during their doctoral work [54, 55], and incorporating advances in MPO techniques from the 2010s.

First things first, let's establish how the A-tensors, that represent any given state (see equation 3.5), which contain all the information states as these are our building bricks. Each $A^{n_k}_{\alpha_k \alpha_{k+1}}$ is a rank-3 tensor, but as we want to keep our library as bear bones as possible, we'll not use a tensor-ready library, just to get a better grasp of what they are doing behind those black boxes called contraction functions, instead, we've flattened those tensors (originally $\mathbb{C}^{m\times M\times M}$ where m is the physical dimension and M is the MPS bond dimension aka the amount of entanglement the state holds) as matrices, as shown in Fig. 3.13. And that being one of the A's for a given

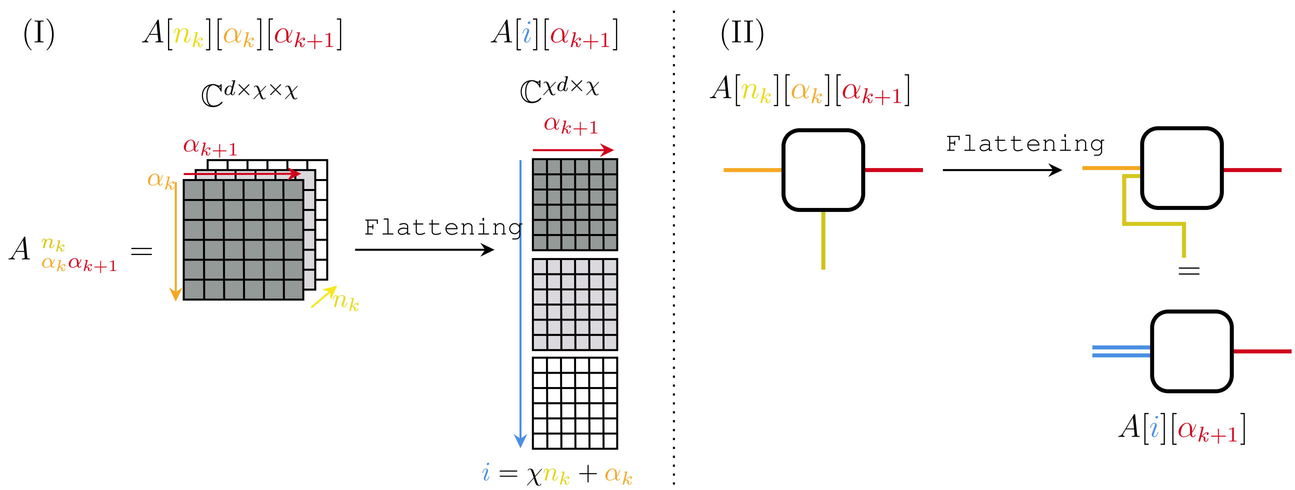


Figure 3.13: (I) Flattening of a rank-3 tensor $A^{[k]}$ into a matrix by grouping (n_k, α_k), while preserving the indices for subsequent contractions. (II) Flattening procedure represented in Penrose notation, where the merging of two lines symbolizes the grouping process, without losing the original indices.

lattice site, we construct the whole coefficient that describes the state as in the following matrix form,

$$\mathcal{A} = \begin{bmatrix} A^{|0\rangle_1} & A^{|0\rangle_2} & \cdots & A^{|0\rangle_L} \\ A^{|1\rangle_1} & A^{|1\rangle_2} & \cdots & A^{|1\rangle_L} \\ \vdots & \vdots & \ddots & \vdots \\ A^{|m\rangle_1} & A^{|m\rangle_2} & \cdots & A^{|m\rangle_L} \end{bmatrix}.$$

There are a few contraction protocols we've come up with which are the heart of this project, such as internal products, expected values or the aplication of an operator on a state. It's important to remember that the $\mathcal{A}$ is a matrices array, turning out to be nothing else than a complex matrix with dimensions $(Mm \times ML)$, that works only as a place holder for our state.

3.6.1 Internal Product

Let us start with the internal product operation as the first protocol to introduce, in order to do so the reader might wonder about the corresponding $\mathcal{A}^\dagger$, for the Matrix Product Bra. It's really straight forward, as of now, our $\mathcal{A}$ is composed by m matrices with $M \times M$ dimensions, to be able to do the dot product between $\mathcal{A}$ and $\mathcal{A}^\dagger$ and get a result that yields a result coherent with something such as $\langle \phi | \psi \rangle$, see Fig. 3.15(I) What we do is to reshape each A_k in $mM \times M$ to $m \times M^2$ as a row array sketched in Fig 3.14. After reshaping the tensors, A_k is transposed and conjugated, then do the matrix multiplication between state and its corresponding dual, already well known, see Fig. 3.15. The result of this is described in sections above as a transfer matrix, but to be able to structure it in a way that will facilitate the Trace of itself, it has to be reshaped in $M \times M$ blocks, that'll correspond to each m that was lost after contracting A and $A^\dagger$, see Fig. 3.16 (II).

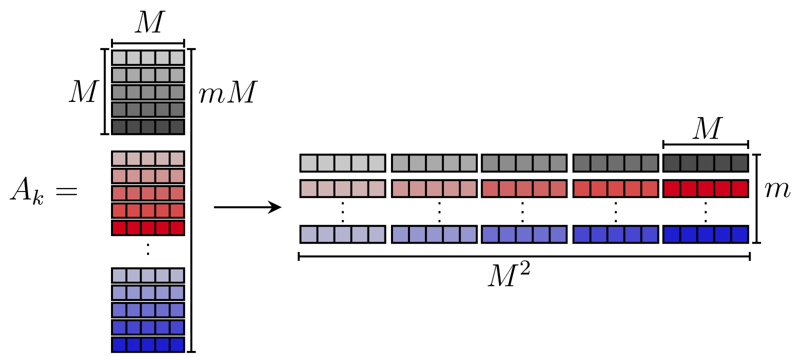


Figure 3.14: Reshaping of the A_k tensor from dimensions $(mM \times M)$ to $(m \times M^2)$ in preparation for contraction with its dual.

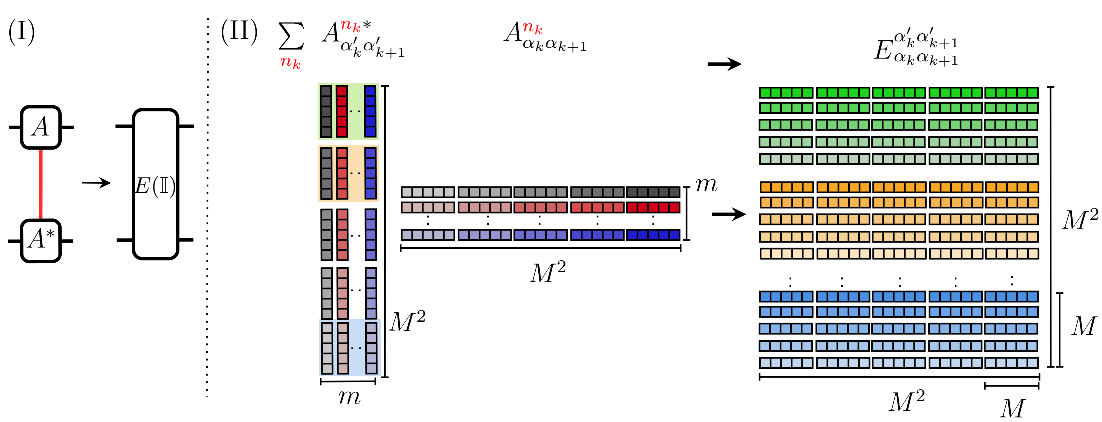


Figure 3.15: (I) Penrose notation for the contraction of the physical index n_k of an A-tensor with its dual, resulting in a transfer matrix whose operator corresponds to the identity. (II) Matrix reshaping and dot product construction of the transfer matrix: both A and $A^\dagger$ are reshaped to form the intermediate representation shown, where the four indices $\{\alpha\}$ correspond to the MPS bond dimensions. (M=5)

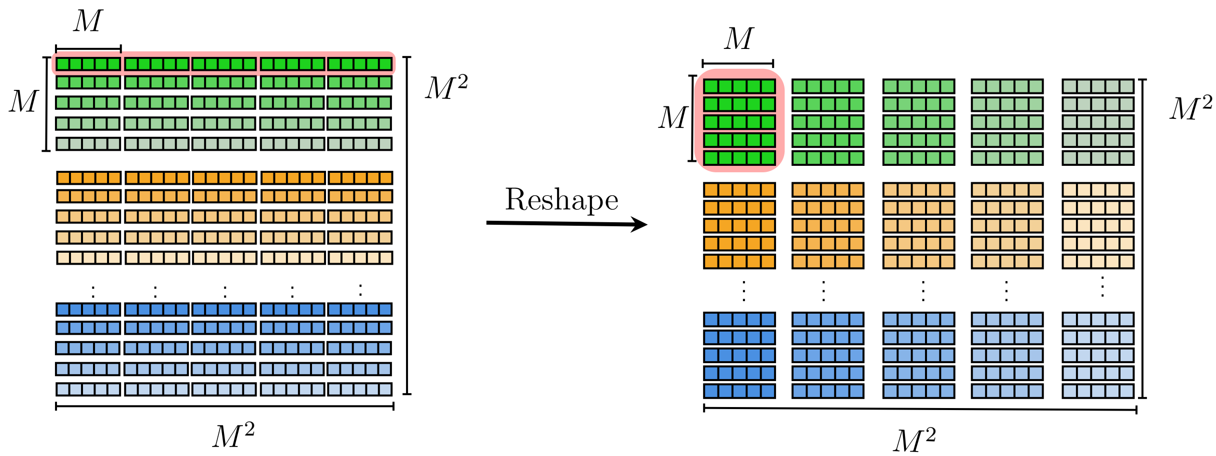


Figure 3.16: Final representation of the transfer matrix, reorganized into $(M \times M)$ blocks rather than $(1 \times M^2)$ rows, which facilitates the trace operation. (M=5)

3.6.2 Operator-State Operation

Another fundamental building block in this journey is the application of operators, written in their matrix form, onto the local A tensors. Our objective during this step is contracting the physical indices (n_k), without mixing information stored in the virtual indices of the MPS $(\{\alpha, \alpha'\})$ in the process. To avoid that, the contraction proceeds in two steps, in the simplest situation (an MPO bond dimension $w = 1$, so we only have a single operator), the physical indices n_k, n_k' are first contracted between the operator $O_{n_k' n_k}$ and the two-legged tensor. This produces an intermediate object $B^{n_k}_{\alpha_k \alpha_{k+1}}$ which still carries the auxiliary indices.

In the second step, we contract over the remaining physical index n_k', yielding the effective tensor $E^{\alpha_k \alpha_{k+1}}_{\alpha_k' \alpha_{k+1}'}$. Diagrammatically, this is represented by pulling the operator through the physical leg of the A tensors, leaving behind a four–index object that acts entirely on the virtual degrees of freedom, which we call Transfer Matrix, see both the expressions and their corresponding diagrams in Fig 3.17. This is how the transfer matrix actually looks like in the code so far, it's important to think on how contractions are performed to get the final form of either Figs 3.9, 3.11 or 3.10, as to get a scalar value we should contract neighboring transfer matrices. We have to worry about the borders first, for that we trace over the elements of α_0 and α_0' to reduce the dimension of a transfer matrix from one border and then contract with its neighbor transfer matrix. The process is followed until reaching the opposite side, where we also trace over α_L and α_L' ending with a $(1 \times M^2)$ column vector times a $(M^2 \times 1)$ row vector, a more general MPO bound dimension where

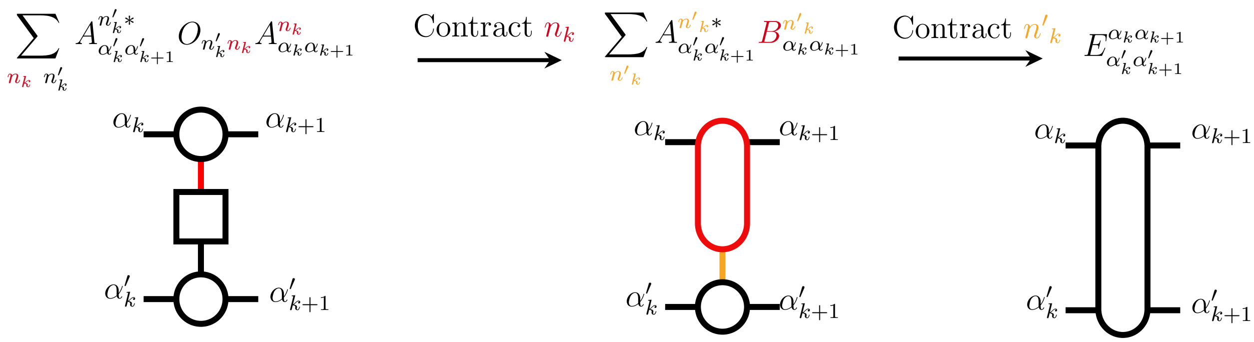


Figure 3.17: Stepwise contraction of tensors in Penrose notation. First, the physical indices n_k are contracted between A, $A^\dagger$, and an operator $O_{n'_k n_k}$. This produces an intermediate tensor B. In the next step, contracting over n_k yields the transfer matrix E, with indices corresponding to the MPS bond dimensions.

$w > 1$ those vectors are actually $(wM \times wM^2)$ column matrix times a $(wM^2 \times wM)$ row matrix.

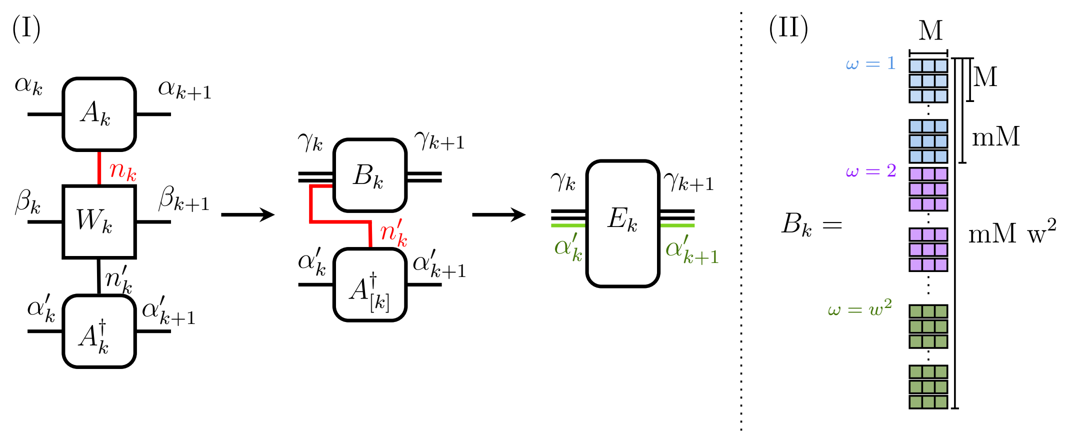


Figure 3.18: (I) Contraction procedure between MPO element in the k-th site ($W^{[k]}$) with it's corresponding MPS tensor A_k to then contract with it's dual $A^\dagger_k$, where the closeness of the legs in each tensor represents how they are group together $\gamma_k = (\alpha_k, \beta_k, n'_k)$. (II) The contraction of the operators in each MPO is done the same way as in Fig. 3.15, but each contraction is stored in w^2 blocks of size $(mM \times M)$, thus bringing the final size of the B_k matrix to be $(mMw^2 \times M)$

Before jumping to the expected value of the transfer matrix shown in 3.17, let's discuss to the aforementioned case, where instead of just local operators we have a proper MPO, i.e $w > 1$. We are already aware of how the transfer matrix is formed from contracting and reshaping A and $A^\dagger$. The only difference here is in the contraction of $W^{n'_k n_k}_{\beta_k \beta_{k+1}}$ with $A^{n_k}_{\alpha_k \alpha_{k+1}}$ as the indices α and β need to be grouped together, therefore increasing the bond dimension from $(mM \times M)$ to $(mMw^2 \times M)$ in our current contraction protocol, that means, a set of w^2 blocks of size $(mM \times M)$. By reusing the operation contraction protocol the original MPS library should be extended for the MPO library construction, so one block-matrix for each operator in the MPO. This contraction protocol in particular has room for improvement, because at least one of those operators is a zero operator in any given MPO, nevertheless, optimization is a perspective of the present work.

After contracting B_k with $A_k^\dagger$ with the same procedure described in Fig. 3.15, but with a matrix size $(M^2 \times M^2 w^2)$, that's reshaped to get the desired block structure shown in Fig. 3.16, the final reshaping consist in regrouping the transfer matrix per block that we desire, to get $(M^2 w \times M^2 w)$ instead, as plotted in Fig. 3.19, this step is crucial due to the fact that organizing leads us to trace easier the transfer matrix, allowing us to finally calculate expected values and also increase performance. For the last step in the expected value, it's important to recall that the first $A^{n_1}_{\alpha_1\alpha_2}$

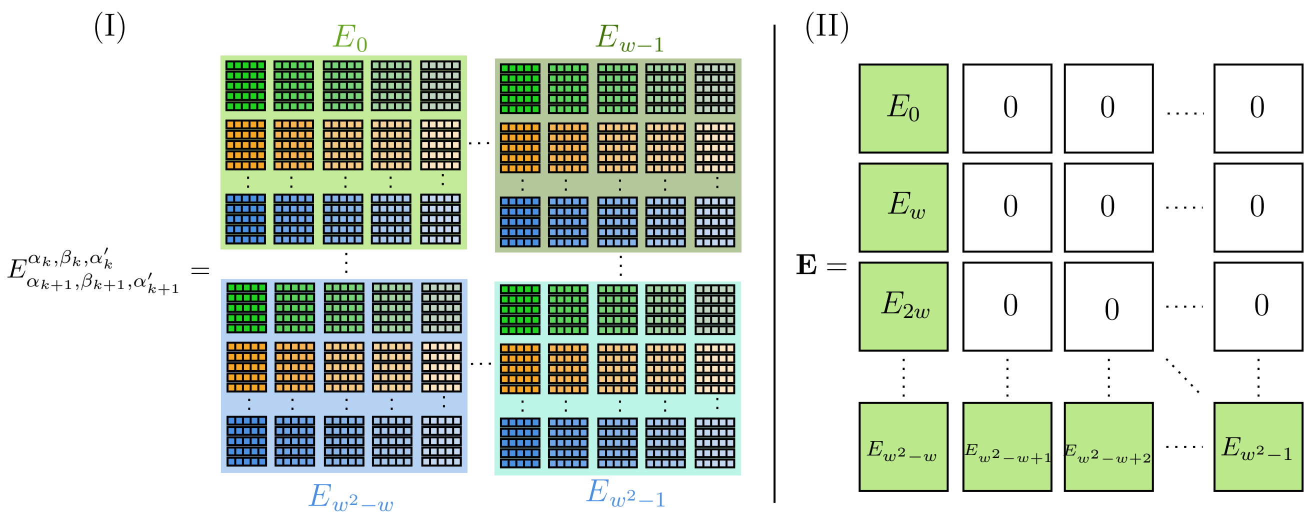


Figure 3.19: (I) Final k-th site transfer matrix, obtained from the contractions in Fig. 3.18 and the reshaping in Fig. 3.16, combining contributions from all MPO operators. (II) Since only the first column and last row contain nonzero blocks, it is enough to store $2w-1$ sub-transfer matrices of size $(M^2 \times M^2)$, reducing the cost from $w^2 M^4$ to $(2w-1)M^4$ and achieving linear scaling in w.

and the last tensor $A^{n_L}_{\alpha_L \alpha_{L+1}}$ in the MPS have what we call dummy legs, $\alpha_1 = \alpha_{L+1} = 1$, therefore some values in our matrix representation vanish to zero, as seen in Fig. 3.20.

This leads to one of the improvements in this work, the **trace over the transfer matrix** used in both MPS and MPO protocols. Many of them were attempted, some more successful than others, but the protocol that gives the proper expected value to every state we've tried so far exploits the fact that the information shaded in red in Fig. 3.20 are just zeros by multiplying the first transfer matrix by a trace vector. In practice this trace vector is implemented with a skinny selection matrix C. Instead of keeping the full identity, C singles out only those entries where the two physical indices of the MPO tensor coincide, i.e. $n'_k = n_k$ (this corresponds to setting $\alpha = \beta$ in the block-matrix notation). Each column of C corresponds to one of these diagonal components, while the MPO bond indices β_k, β_{k+1} are carried along as usual. In this way, when the transfer matrices act on C, only the diagonal physical contributions are propagated through the chain.

The final summation then contracts these propagated columns with the left boundary, which enforces the remaining trace over the physical degrees of freedom and produces the measurement. In the code the index γ is used to label the row blocks of the transfer matrix, while γ' labels the column blocks. Both are composite labels obtained by flattening the two MPO bond indices (β_k, β_{k+1}) into a single block index, so that $\gamma \leftrightarrow (\beta_k, \beta_{k+1})$ on the row side and $\gamma' \leftrightarrow (\beta'_k, \beta'_{k+1})$ on the column side. These should not be confused with the MPS bond indices (α_k, α_{k+1}); they belong purely to the MPO structure and run over the w possible bond sectors. The selection matrix C has dimensions $(wM^2) \times (wM)$, with $N = wM^2$ rows and $K = wM$ columns. Each column is indexed by $j = \gamma' M + \alpha$, while each row corresponds to a flattened index $\mathit{global\_col} = \gamma M^2 + \alpha M + \alpha'$. The nonzero entries are placed exactly when $\alpha' = \alpha$, i.e. on the diagonal of the doubled MPS bond space, so that

$$C_{\mathrm{global\_col},\, j} = \delta_{\gamma,\gamma'}\, \delta_{\alpha',\alpha}.$$

In other words, C projects onto the diagonal components ($\alpha' = \alpha$) inside each MPO block, and in this way implements the partial trace over the MPS auxiliary indices. Now that all measuring protocols have been described, we are ready to jump over the characterization of MPS and test all

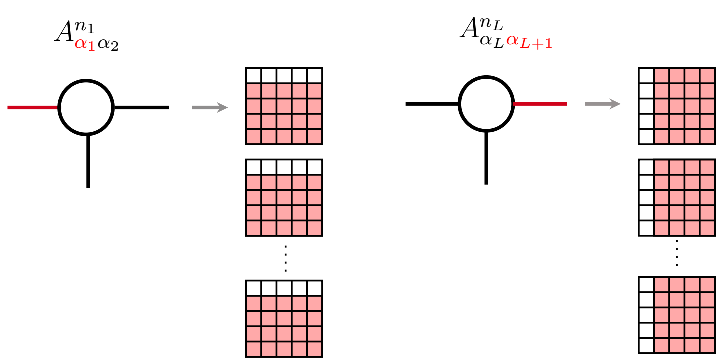


Figure 3.20: Zero-padding of the first and last A-tensors, indicated in red, ensuring that the full TN retains the correct structure during the contraction process.

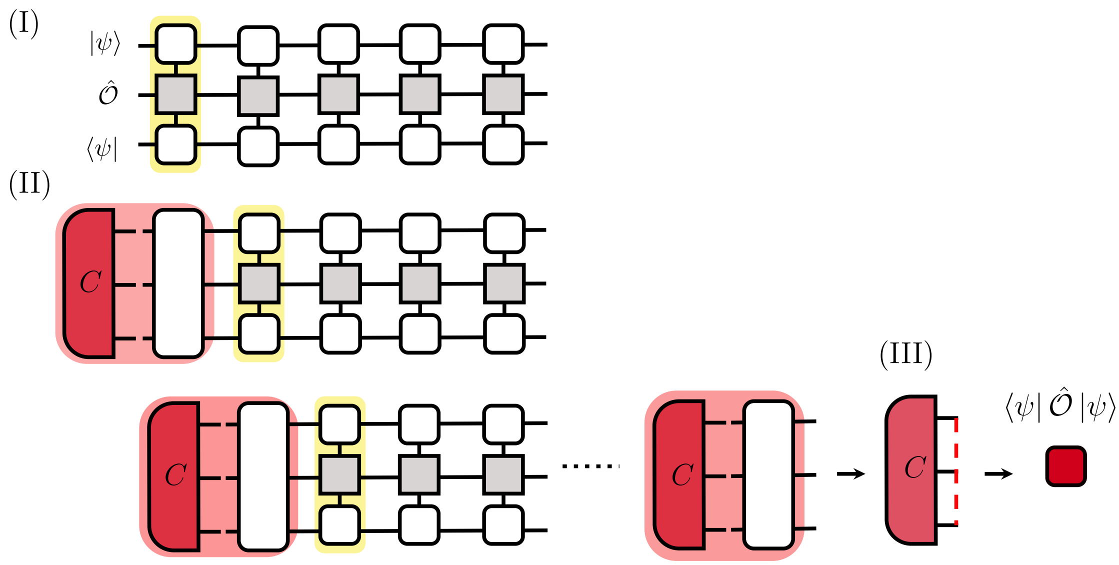


Figure 3.21: Final contraction of the TN: (I) form the transfer matrix by contracting the operator with the MPS and its conjugate; (II) sequentially contract the transfer matrices with the boundary vector C; (III) after all contractions, trace over C to obtain the expectation value.

the developed MPO-library, which is actually the trask we face on in Chapter 4.

Chapter 4

Results

The broad objective of this project is to open a new numerical branch in the Theoretical Solid State Physics research group. First, lets recall the research objectives and the strategy followed, namely the use of tensor network algorithms applied to the spin-1 Bose-Hubbard model with an external Zeeman field. We then structure the results in three main parts, corresponding to the objectives introduced in Chapter 1.

4.1 Testing MPO developed library

The physical basis used for the calculation of expected values is the magnetic basis, for the details regarding the order of the basis as well as the way each operator was constructer the reader should head to Appendix B.2. This form leads us with extended local operators where solely local basis needs to be considered per lattice site, and the operators written as product of local-operators, allows us to perform any local characterization. In the previous method, implemented in what we

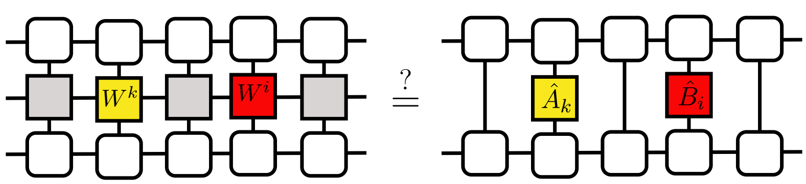


Figure 4.1: Comparison of two strategies for evaluating correlation functions: (right) direct contraction within an MPS framework and (left) MPO representation of the correlator.

now call Approach A, correlation functions were computed as illustrated on the right-hand side of Fig. 4.1: one operator acts on the k-th site and another on the $i = k + \Delta$ site, with all tensors in between fully contracted. The ground states used throughout this work were obtained via a variational DMRG procedure using this same Approach A library.

By contrast, the more recent Approach B allows the representation shown on the left-hand side of Fig. 4.1, in which the entire two-point correlator is encoded as an MPO (see Eq. 4.1). Our goal in this chapter is to verify that both approaches yield consistent and compatible results.

$$\hat{A}_k \hat{B}_i = \prod_{j=1}^{L} W^{[j]}, \qquad W^{[j]} = \begin{cases} \begin{pmatrix} \hat{1} & \hat{0} & \hat{0} \\ \hat{B}_j & \hat{0} & \hat{0} \\ \hat{0} & \hat{A}_j & \hat{1} \end{pmatrix} & j \in \{k, i\} \\ \begin{pmatrix} \hat{1} & \hat{0} & \hat{0} \\ \hat{0} & \hat{1} & \hat{0} \\ \hat{0} & \hat{0} & \hat{1} \end{pmatrix} & \text{otherwise.} \end{cases} \tag{4.1}$$

Finally, we show the general scope of the present work by allowing the increment of the local basis, where we consider a bosonic system with up to two bosons per site, which leads us to a basis with $m = 16$ elements (See Appendix B.6).

4.2 Physics within the Bose-Hubbard Model

To characterize the physics of the BH model Eq. 2.8, we calculate the expectation value of a set of local observables and correlation functions in both the Mott insulating regime via the effective generalized Heisenberg model in Eq. 4.2 (see section 4.2.1) and the superfluid regime. The local quantities include the total number of particles $\langle \hat{N} \rangle$, the particle number per magnetic projection $\langle \hat{n}_{-1} \rangle$, $\langle \hat{n}_0 \rangle$, and $\langle \hat{n}_1 \rangle$, as well as the magnetization $\langle \hat{S}^z \rangle$.

In addition, we analyze two-point correlations, namely $\langle \hat{S}^z_k \hat{S}^z_{k+\Delta} \rangle$, $\langle \hat{S}^+_k \hat{S}^-_{k+\Delta} \rangle$, and $\langle \hat{S}^{+2}_k \hat{S}^{-2}_{k+\Delta} \rangle$, across both regimes. Finally, we also compute the one body correlator $\langle \hat{b}^\dagger_{k,\sigma} \hat{b}_{k+\Delta,\sigma} \rangle$, which provides direct insight into phase coherence in the superfluid phase.

4.2.1 Assesing the MPO protocols with Mott insulating MPS

Our numerical results proceed in two stages. First, we compare the previous MPS implementation with the current one based on MPOs, establishing consistency between both approaches. Once the measurement protocols are validated, we characterize the states exclusively within the MPO framework. We begin with the Mott-insulating regime of the Bose-Hubbard model at unit filling per site, as mentioned in Section 2.2.

In close analogy to the fermionic Hubbard model, whose low-energy sector in the strong-coupling regime is famously described by an effective spin-$\frac{1}{2}$ Heisenberg antiferromagnet[56]; the Mott insulating phase of the spin-1 Bose-Hubbard model at unit filling can likewise be mapped onto an effective spin Hamiltonian. In this limit, charge (density) fluctuations are gapped, and virtual tunneling processes between neighbouring sites generate purely spin-exchange interactions. The resulting low-energy description is the bilinear-biquadratic spin-1 Heisenberg model [57, 58], shown in Eq. 4.2.

Within this mapping, the on-site Hilbert space reduces from the full Bose-Hubbard basis of dimension $m = 16$, which includes all allowed occupation and spin configurations, to the three spin-1 states $\{|m=-1\rangle, |0\rangle, |+1\rangle\}$, i.e. $m = 3$. While both descriptions capture the same spin physics in the Mott regime, we employ this reduced spin-1 framework for computational efficiency and for a more direct comparison with the standard bilinear-biquadratic phase diagram.

$$\hat{H}_{BBH} = \sum_i \cos\theta\, \mathbf{S}_i \cdot \mathbf{S}_{i+1} + \sin\theta\, (\mathbf{S}_i \cdot \mathbf{S}_{i+1})^2 + D \sum_i \left(\hat{S}_i^z\right)^2 . \tag{4.2}$$

It's worth noting that this regime of the BH has many phases [58], the reader is addressed to Fig. 4.2 and [59] for further analysis of the Mott-insulator phase diagram. The ones whose ground states are here characterized can be found in Table 4.1. We do not discuss in detail the physical description of the considered states, instead we focus on the observable measurements on the states.

In Fig. 4.3, we present the site-resolved populations of the three spin projections across the Large-D phase. The behavior is explicit in both the Antiferromagnetic Large D as well as the Ferromagnetic phases (see Fig. 4.3) the dominant population resides in the $S = 0$ projection, a

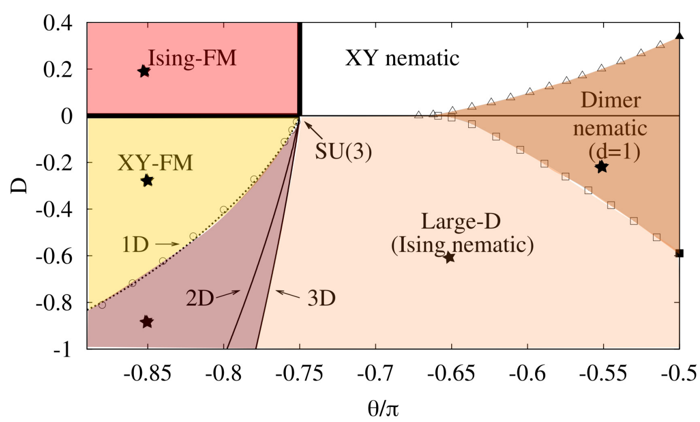


Figure 4.2: Different phases within the Mott-Insulator regime, highlighting the ones analyzed. Adapted from [59].

MI Phase	D	θ/π
Large-D Antiferro	-0.6	-0.650
XY Ferro	-0.3	-0.850
Large-D Ferro	-0.9	-0.850
Dimer nematic	-0.2	-0.550

Table 4.1: Parameter values for different phases within the effective Bilinear Biquadratic Heisenberg Hamiltonian that describes the Mott-Insulator regime.

hallmark of the nematic character of this phase. Despite these differences, the phases share an important feature: the total population per site is fixed to one particle, reinforcing the consistency of the Mott regime. Furthermore, the results obtained with the MPO-library (see Fig. 4.3) are in perfect agreement with those from the earlier MPS-library, confirming the correctness of the MPO implementation at larger bond dimensions and showing that the previous protocol is now encompassed within the broader MPO framework. Moving on to Fig. 4.4, in the realizations of the XY-FM phase, we observe that the magnetic field $Q = -0.3$ cause a redistribution of populations, the $S = \pm 1$ projections are still mostly occupied as opposed to while the $S = 0$ population.

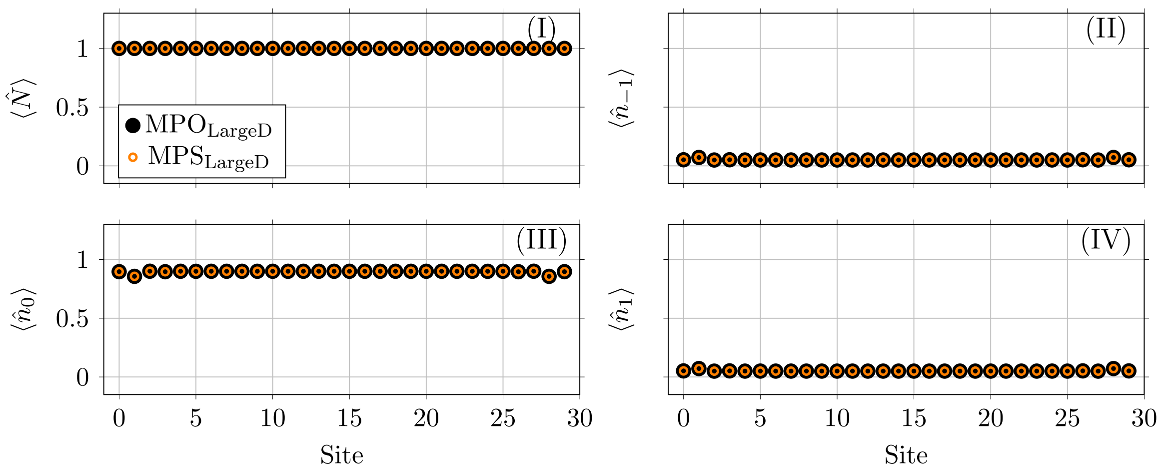


Figure 4.3: Comparison of expectation value of the density of states operator per site between MPO (black circle) and MPS (red hollow circle) in the Large-D AFM phase in the MI regime, where the observables are (I) number of particles per site, and the profile magnetic projections per site (II) $m_F = -1$ (III) $m_F = 0$ (IV) $m_F = 1$. And $M = 30, \quad L = 30$.

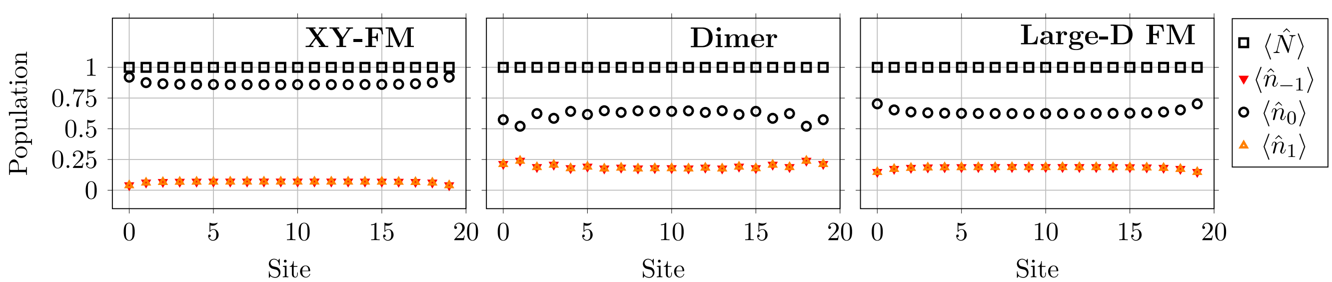


Figure 4.4: Expectation value of the density of states operator per site in different phases within the MI regime (See Table 4.1)with a total number of sites of L = 20. No comparison to MPS was made due to the nice performance shown in Fig. 4.3.

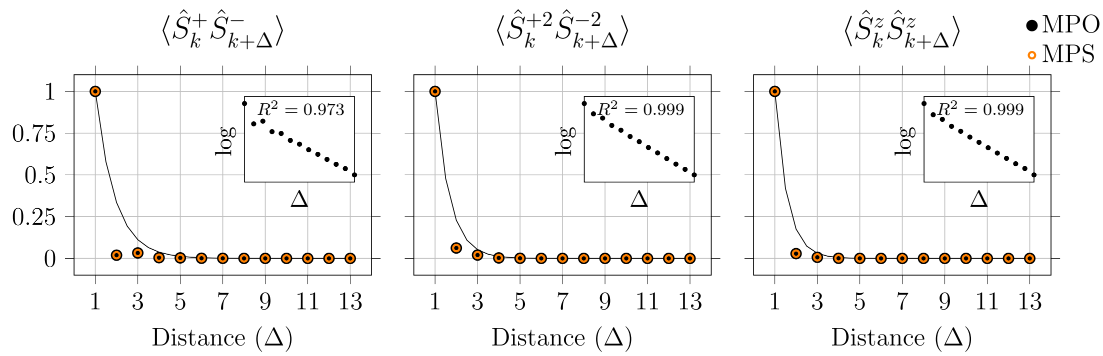


Figure 4.5: Three kinds of two-point spin-correlators inside the Antiferromagnetic Large-D phase in the MI regime, plotting the correlation against the neighbour distance (Δ) from the operator centered in $k = 2$ and its natural logarithm in each inset plot. The parameters can be found in Table 4.1 and additionally $M = 30, \quad L = 30$.

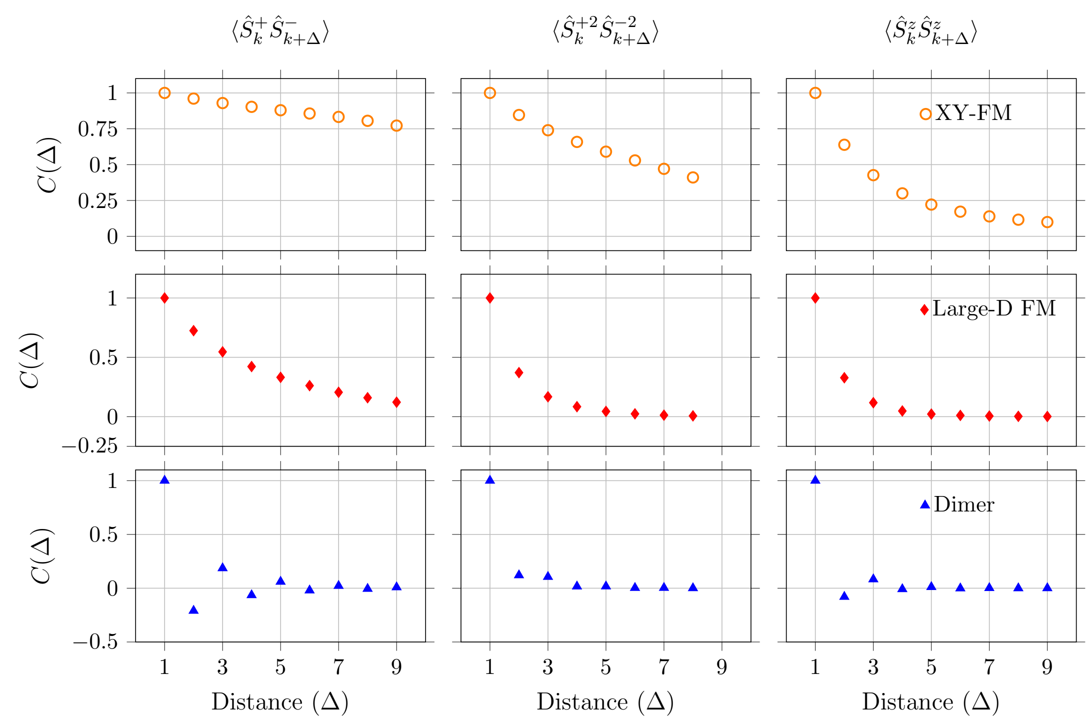


Figure 4.6: Three types of two-point spin-spin correlations ($C(\Delta)$), two longitudinal and one transversal to the lattice direction inside several phases described in Table 4.1. The parameters in common are $M = 18, \quad L = 20$.

In Fig. 4.5, the transverse and longitudinal spin–spin correlators decay exponentially, as evident from the linear trends in the semi-logarithmic insets. Fits to $C(\Delta) = \exp[-(\Delta - 1)/\xi]$ yield $\xi \approx 0.914$, 0.677, and 0.575, respectively, consistent with a gapped magnetic phase in which a smaller ξ implies faster spatial decorrelation. More generally, ξ is the intrinsic length scale of correlations, it is finite in gapped regimes, grows upon approaching a quantum critical point, and diverges at criticality where correlations become algebraic. Computationally, ξ sets the size and bond-dimension requirements, to access bulk behavior one needs windows and system sizes $L \gg \xi$ to suppress boundary effects. In tensor-network simulations the effective ξ increases with bond dimension M (See Fig. 4.11), so gapped phases with small ξ are captured with modest χ, whereas near criticality larger χ is required and errors scale with the finite entanglement length.

Now switching to Fig 4.6 the data also supports the expected phase physics. In the XY–FM regime the longitudinal correlator $\langle \hat{S}_k^+ \hat{S}_{k+\Delta}^- \rangle$ and $\langle (\hat{S}_k^+)^2 (\hat{S}_{k+\Delta}^-)^2 \rangle$ decrease monotonically, the first being dominant and decaying slowly with distance-consistent with quasi long-range decay, while $\langle \hat{S}_k^z \hat{S}_{k+\Delta}^z \rangle$ remains small and decrease rapidly in comparison, indicating stronger suppression on the transversal fluctuations than on the longitudinal fluctuations. Moving from XY–FM to Large-D FM the overall magnitudes drop, consistent with stronger easy–plane anisotropy. In the Dimer regime all channels are short ranged and rapidly vanish with Δ, characteristic of a gapped, nematic state.

4.2.2 Assesing the MPO protocols with Superfluid MPS

Now that we have gone over the Mott-Insulator regime, we turn to the superfluid phase of the Bose–Hubbard model. For this ground state we use $t = 1$ (energy unit), $\mu = 1$, $B = 0.5$, and interactions $g_0 = 0.5$ with $g_2 = 2g_0$ (ferromagnetic) or $g_2 = 0.5g_0$ (antiferromagnetic), on a system of size $L = 8$ with bond dimension $M = 18$. A notable reduction in the size (8 sites) of the network is evident due to the extreme increase in the local basis (16 states), all in order to be able to calculate the ground state of the system in a reasonable computing time (of the order of 30 hours due to the amount of internal degrees of freedom, which scales as 16^L) by means of the variational MPS.

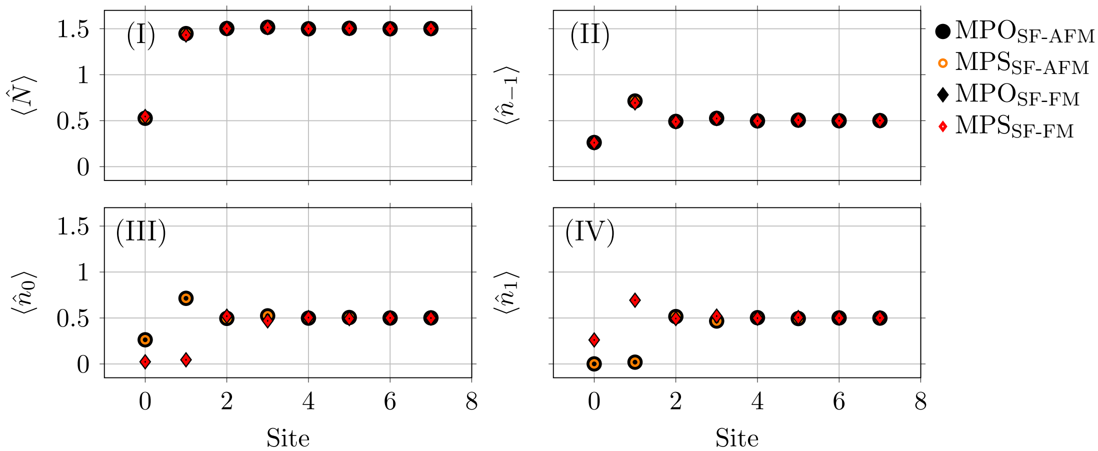


Figure 4.7: Particle density per site for the superfluid regime in the Bose Hubbard model in a $L = 8$ lattice, $M = 18$, $m = 16$. The observed skew in the density reflects incomplete convergence of the variational state, which has not yet reached the fully symmetric ground state.

As shown in Fig. 4.7, the site-resolved densities by magnetic projection display the fingerprint of the superfluid, the variance of the occupation per site is not zero. The finite hopping amplitude delocalizes particles across the lattice, confirming the superfluid character of the ground state. This is further supported by the variance of the number of particles per site (see Fig. 4.8), where if it's non-zero we are certain that the system does not belong to the Mott-Insulator regime, regardless of how further measurements as correlations may look like.

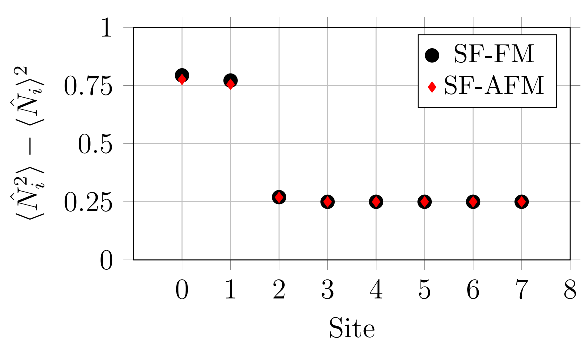


Figure 4.8: Variance of the number of particles per site in both the ferromagnetic as well as the antiferromagnetic phases of the superfluid regime, showcasing that the ground state under consideration does not belong to the Mott-Insulating regime.

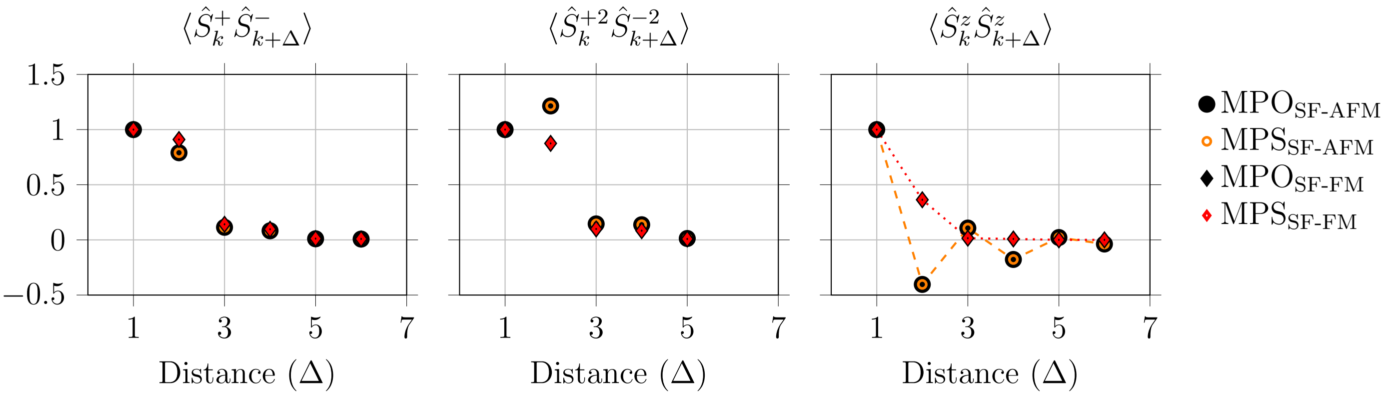


Figure 4.9: Longitudinal and transversal two-point correlators for ferromagnetic and antiferromagnetic states belonging to the SF regime. Measured from $k = 2$

While a non-integer occupation per site rules out the Mott-insulator, it does not, by itself, establish superfluidity. Inspecting the populations in Fig. 4.7, the profiles relax to a bulk plateau beyond roughly the third site, suggesting that boundary effects dominate near the edges and that an effective window of about $L \simeq 6$ sites remains for meaningful correlation measurements, and even so, with the variance we can make sure that this is indeed the case. Although it is safe to admit, this system is too short to draw definitive conclusions about the ground state's character. Let us remember that our main task is to compare the MPO-protocol (Approach B) with the previously implemented MPS-library (Approach A) by characterizing meaningful states, in this occasion, states made up by a large set of 16 states in the local basis.

Identifying a superfluid behaviour would require correlation diagnostics together with finite-size scaling, ideally on longer chains, which is not the case with the measurements in Fig. 4.9. Even though the system has few sites, a remarkable behaviour can be seen in the $\hat{S}_z$ correlations in Fig. 4.9, these imply that the states are not entirely skewed; the antiferromagnetic decays in a zig zag pattern, and the ferromagnetic decays monotonically.

4.3 Contrasting M and L in the SF regime

Let us first recall that M is the entanglement parameter, which measures the amount of entanglement that the system may exhibit under study. On the other hand, L is the lattice size, which determines whether or not bulk properties, edge effects, among others can be studied. For the time available at our hands we could see converging a grand total of 217 states, with values going for $M \in [8, 18]$

and $L \in [12, 60]$.

As mentioned earlier, one of the challenges we faced is the computational time the variational-MPS takes for the states to converge[1]. A considerable amount of states has been obtained with the paremeter set as: $t = 1.0$, $B = 2.0$, $g_0 = 3.0$ and $g_2 = 2g_0$. The most relevant correlators here analyzed are, the one body correlator $C(\Delta) = \left\langle \hat{b}^\dagger_{i,\sigma} \hat{b}_{i+\Delta,\sigma} \right\rangle$ and the axis and plan spin-spin correlators.
The main difference between the Mott-Insulator and the superfluid regimes, besides the variance of the number of particles per site, can be boiled down to coherence, the correlations in the superfluid decay algebraically, $C(\Delta) \approx \Delta^{-r}$, while the Mott Insulator's exponentially, $C(\Delta) \approx e^{-\Delta/\xi}$. And although in Fig 4.7 the total density shows a non-commensurate number of particles per site, confirmed by a finite variance (see Fig. 4.8), the correlations for such a few-sited chain in Fig 4.10 are decaying exponentially, as shown in the inset of each plot, regardless of M or L. This is actually precisely why in Fig 4.11 a heat map of the correlation length ξ is shown for each value of L and M. One important highlight is that only even chains are taken into consideration, due to the fact that we would like some mirror symmetries to be preserved in the system. Continuing with the unexpected behavior of the superfluid phase we find ourselves in, we were curious on what would happen if we increased the length of the whole chain, would these correlation lengths end up converging somewhere? and if so, is there also an optimal value of M we can use to avoid overclocking our clusters to get more data that could be recollected with a smart and informed selection of parameters? By definition, in the fundamental theory of MPS, we can't choose M to be lower than the actual physical dimension, $m = 16$ in the superfluid case. Nevertheless, the behaviour found in these systems, although without the amount of entanglement necessary, still presented in the correlations as in the examples analyzed in Section 4.2.2.

Computational time was a major constraint but still we focused on the one-body correlator $\langle \hat{b}^\dagger_{i,\sigma} \hat{b}_{i+\Delta,\sigma} \rangle$, since it constitutes a key distinction between the superfluid and Mott-insulator regimes. Interestingly, Fig. 4.11 shows correlation lengths in the superfluid regime an order of magnitude larger than those in the Mott-insulator regime, reflecting the much slower decay of correlations in this case meaning that regardless of the lattice's size, short or long, the correlation length allows us to capture physical differences between these two regimes supported by the Bose-Hubbard model. And more importantly, the MPO-protocol implemented in the present work is capable of performing the correlator calculations that allow us to capture this physics.

[1] But thanks to the internship my partner A. Galindo got at OIST, we could use their cluster, which made what would have been months of computing time in just a couple of weeks.

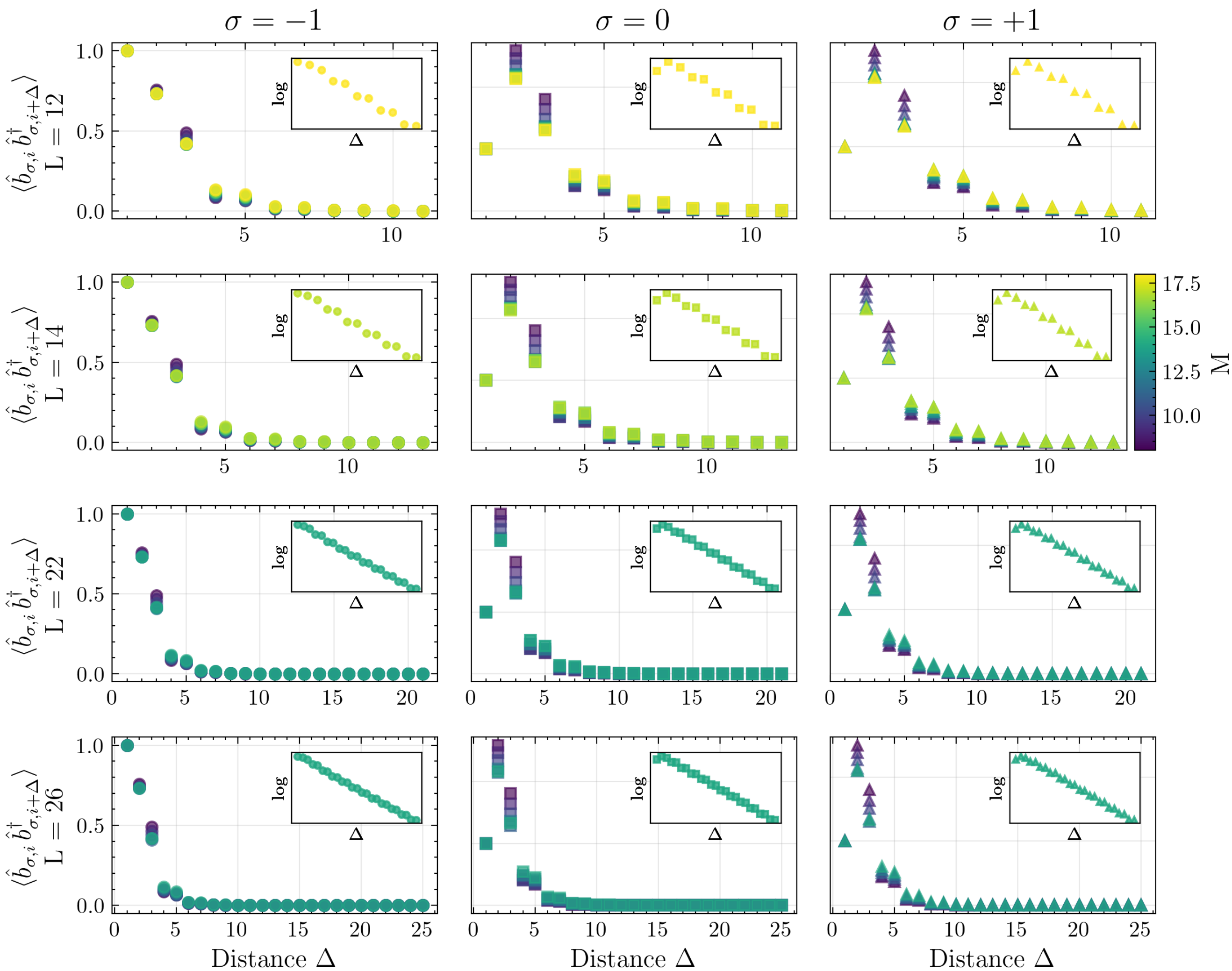


Figure 4.10: Nine kinds of one-body correlators inside the superfluid Ferromagnetic phase, plotted against the neighbour distance (Δ) for different bond dimensions M. Each column corresponds to a different spin projection $\sigma = -1, 0, +1$, while each row shows the correlators obtained from distinct chain sizes. Inset panels display the same data in logarithmic scale.

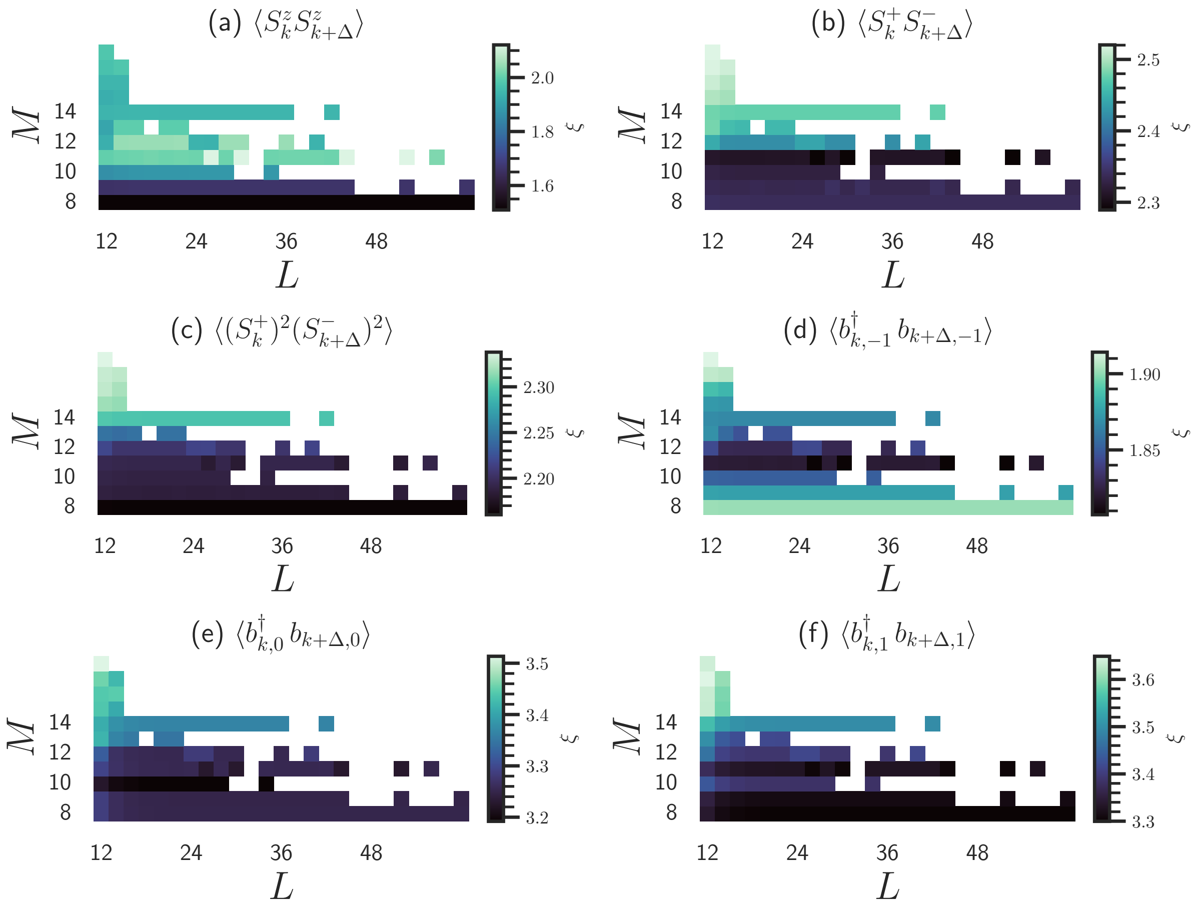


Figure 4.11: Correlation-length heatmaps for different two-point correlators inside the superfluid Ferromagnetic phase. Each panel corresponds to the operator indicated in its label, with the horizontal axis showing the lattice length L, the vertical axis the MPS bond dimension M, and the color scale representing the correlation length ξ.

By increasing the number of sites, even allowing low entanglement ($M < m = 16$), Fig. 4.11 highlights the values of M for which the largest range of L converged, revealing a pattern in the behavior of the correlation lengths. Beyond confirming their slower decay compared to the Mott-insulator regime, the data suggest that from $L = 19$ the correlation lengths converge to the same value, as indicated in Fig. 4.12, indicating possible candidates for optimal chain lengths to study bulk properties in the future. Overall the correlation length provides valuable insight into both the numerical method (decay rates, the influence of open boundary conditions) and the physical question of the minimal system size required to capture bulk behavior. This information, key to future research in the Theoretical Solid State Physics group, has been obtained through the use of the implemented MPO

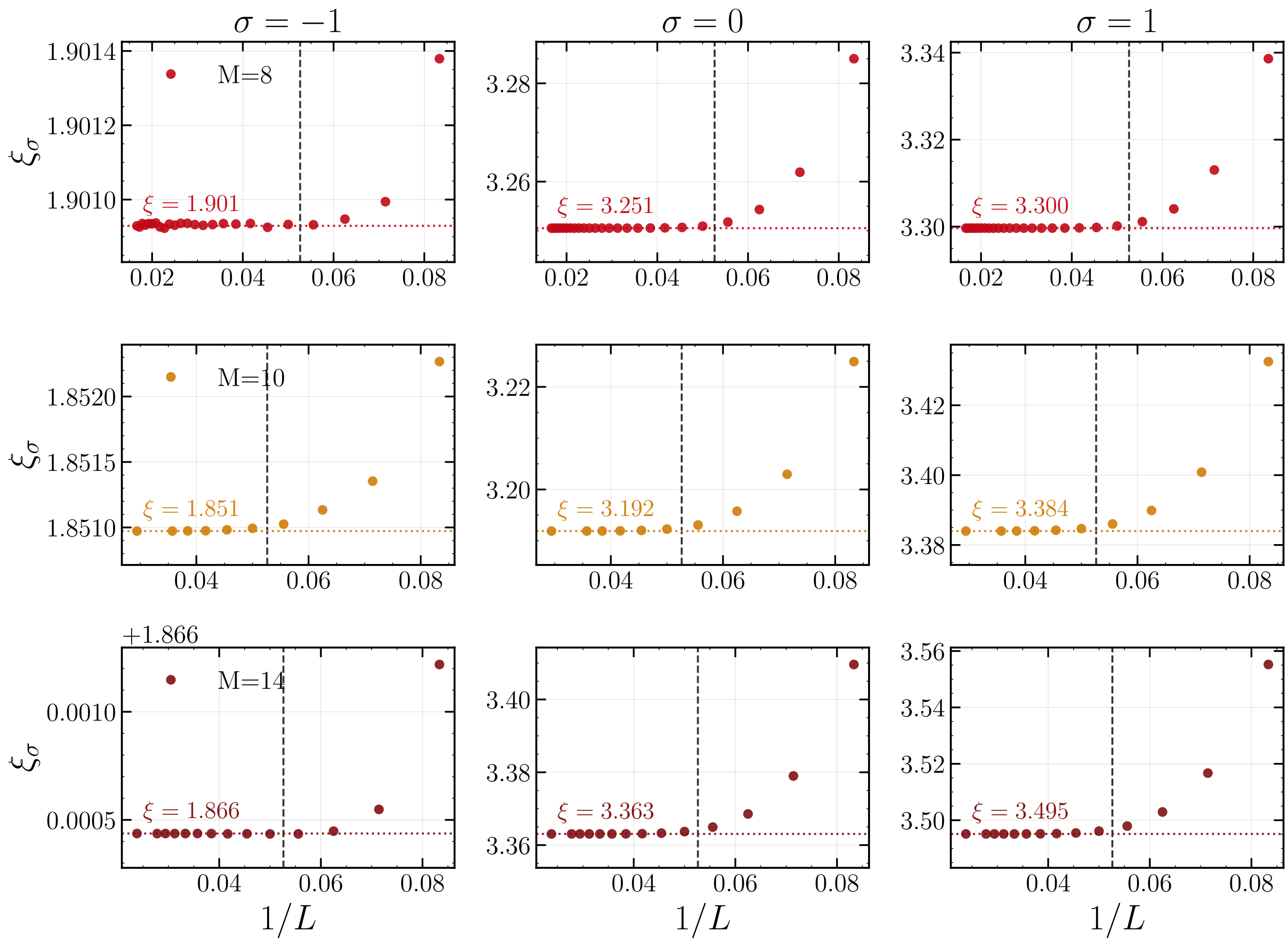


Figure 4.12: Correlation length ξ_σ in the superfluid Ferromagnetic (SF-FM) phase, plotted as a function of the inverse lattice size $1/L$. The columns correspond to spin indices $\sigma = -1, 0, 1$, while the rows correspond to different MPS bond dimensions $M = 8, 10, 14$. Horizontal dashed lines indicate the extrapolated values of ξ_σ at the largest accessible L, which are also annotated inside each panel. A vertical dashed line at $1/L = 1/20$ is included for reference.

Chapter 5

Conclusions and Outlook

5.0 Conclusions

In this work, after having introduced the field in Chapter 1, we presented a thorough development and description of the Bose Hubbard model in Chapter 2.2.1. Particular attention was given to its formulation from first to second quantization, including the addition of the spin-1 degree of freedom. This allowed us to explicitly describe the different types of interactions, distinguishing between spin-preserving and spin-changing collisions. Furthermore, we introduced an external linear Zeeman field and demonstrated how, through its contribution to the hopping term, it lifts degeneracies among the different magnetic projections in momentum space.

Chapter 3 is the heart, we revisited an MPS-based library originally developed more than a decade ago, at a time when MPO methods were not yet established. We not only generalized its scope, but also introduced a more efficient strategy to optimize the trace of the mps transfer matrix, rather than iterating over entries that invariably vanish by construction, we avoided unnecessary operations, significantly improving computational performance. The previous transfer matrix, limited to a rank-4 structure $E^{\alpha_i \alpha_{i+1}}_{\alpha'_i \alpha'_{i+1}}$, has been generalized to a rank-6 tensor $E^{\alpha_i \beta_i \alpha'_i}_{\alpha_{i+1} \beta_{i+1} \alpha'_{i+1}}$, where MPO bonds are explicitly included. This extension is achieved without resorting to tensor-specific libraries, relying instead on systematic flattening and contraction of matrices. It is worth highlighting that the generalized framework reproduces the results of the original MPS-based implementation, while offering the additional advantage of representing both local and non-local operators in matrix product form which in turn make measurements a highly parallelizable implementation.

One of the main benefits of this approach is the treatment of more than one-site operators such as the Hamiltonian due to, for instance, the hopping term. In earlier implementations, any of such kind of operators had to be tensor-product extended from one site to two and then to four in order to exploit translational symmetry and extract the ground-state energy. In contrast, the MPO formalism allows non-local operators to be expressed far more naturally. On top of that, what scaled as m^2 now scales as mw, where m is the physical dimension and w the MPO bond dimension. Since w is typically much smaller than m in high spin systems, this improvement constitutes a considerable computational saver and avoids the risk of incorrect measurements caused by cumbersome operator extensions. This not only saves enormous computational resources, but also makes it feasible to study larger and more complex systems with confidence in the reliability of the results.

Finally, the work has been carefully documented, preserving the original library while providing a fully modernized MPO-based toolbox. This effort re-establishes a numerical tradition in the Theoretical Solid State Physics research group of Universidad del Valle, and lays a foundation for future research directions, simulations of interacting systems with non-local couplings, efficient representations of many-body Hamiltonians, and extensions to time evolution or finite-temperature calculations. The development of this MPO library is therefore more than a technical contribution, it is a mid- and long-term investment that ensures the group is equipped with state-of-the-art computational methods, positioning our research to remain competitive and innovative for years to come.

In Chapter 4 we tested the developed MPO-library by starting from ground states previously calculated with the MPS-library and using them to characterize some phases of the Mott-insulating regime of the Bose–Hubbard model. These phases can be described by the bilinear–biquadratic Heisenberg model, and we studied them by measuring the particle density per site and by computing spin–spin correlators, which helped us clarify their behavior. The chosen states for the characterization are not trivial, since they involved long chains of length $L = 20$ and $L = 30$ with entangled phases. Although keeping low amount of basis states, this is the case for those Mott-Insulating states with three magnetic projections plus the vacuum, whose inclusion must be done by construction; making a local basis of four states ($m = 4$). Yet the results matched faithfully those of the earlier MPS-library. In fact, they correspond to the special case $w = 1$, which confirmed the MPO-library as both a continuation of the doctoral work of A. Arguelles and K. Rodriguez [60] and as a new tool for studying one-dimensional gapped Hamiltonians.

The other group of non-trivial ground states in our task of characterization belonging to the Bose–Hubbard model is in the superfluid regime, with a local basis of $m = 16$, a considerably larger set than the needed in the Mott-insulating regime. Here we compared the convergence of the correlation length for one- and two-body correlators with respect to the MPS bond dimension M and the chain length L. Since our goal is to check whether the MPO-protocol is capable of measuring observables that allow us to understand the system in the thermodynamic limit, we found that the bulk value of the correlation length can be reached with as few as 40 sites, this constitutes the minimum system size that provides the ground state itself for having enough sites to capture bulk behavior. We also found that only chains with even length give the correct physical properties, while odd chains break mirror symmetries and fail to reproduce the expected results. A practical challenge still lies upon the convergence of the ground states in the standard MPS-variational library, where in many cases calculations may take days to converge. This explains the gaps in the heatmaps, which we expect to fill in the future as we explore higher values of M and identify an optimal balance between computing time and the ability to capture entanglement-rich physics.

5.0 Perspectives

Some of the richest physics and applications of these computational tools are still ahead. In this era of tensor networks, where quantum computing may have found a serious competitor, it is crucial that groups like ours at Universidad del Valle continue to develop these techniques before the field is dominated entirely by industrial giants.

In the near future, one natural direction is the implementation of already well established tensor network algorithms for ground state search, such as either the variational algorithm [61] or the Imaginary Time Evolving Block Decimation [62] algorithm. Incorporating into our newly-implemented library would represent a significant step forward. For instance, variational-MPS would require expressing time evolution gates as Matrix Product Operators (MPOs), or alternatively constructing an MPO directly from their contraction, followed by truncation via singular value decomposition (SVD). While technically demanding, such a task is achievable and would yield highly rewarding results lowering the lengthly computational time for ground-state calculations.

Another promising avenue is the development of infinite size Matrix Product States (iMPS) [63], which introduce left and right bulk vectors to approximate the thermodynamic limit without simulating hundreds of sites explicitly. This would allow the study of critical systems and long range correlations at a fraction of the computational cost which is highly desired currently in the research group.

Optimization of our tensor contraction schemes also remains a fertile ground for progress. Although efficient contraction orders are well documented in the literature, our current implementation is far from optimal. A dedicated effort to encapsulate tensors into a flexible object or class based framework, or migrating parts of the library to a robust environment such as iTensor, would open the door to both performance improvements and easier algorithmic extensions.

Beyond algorithmic efficiency, there is also an exciting conceptual horizon. Tensor networks have recently been used to reproduce and explore quantum error correcting codes [64], a cornerstone of fault tolerant quantum computation. Venturing into this direction could open an entirely new research line within the Universidad del Valle, connecting condensed matter, quantum information, and computational physics. Other natural extensions include the implementation of real time evolution algorithms to study dynamics and quenches, finite temperature methods based on purification or minimally entangled typical thermal states, higher dimensional tensor networks such as projected entangled pair states for exploring two dimensional spinor boson systems, and systematic benchmarking of our in house code against established libraries.

In short, there is an ocean of possibilities. What has been built here should serve as a foundation for newcomers to expand upon, with both technical courage and scientific curiosity. If embraced with passion, this line of work can consolidate a strong local research tradition, positioning Universidad del Valle at the forefront of tensor network based quantum simulation.

Appendix A

Appendix A

A.1 From first to second quantization

Turning to second quantization a better formalism to deal with this many-body system. Rather than individual particle wavefunctions as in first quantization, the fundamental objects here are field operators [29] (see equation A.1).

$$\hat{\Psi}(x) = \sum_i W_i(x)\hat{b}_i, \qquad \hat{\Psi}^\dagger(x) = \sum_i W_i^*(x)\hat{b}_i^\dagger. \tag{A.1}$$

As is known from solid-state theory [65] the single-particle energy eigenstates of the Hamiltonian are Bloch functions $\psi_{k,n}(x)$, we'll be working in an approximation where the temperature is low enough such that the lattice potential is strong enough where the band-gaps are large and well separated energetically, thus, only the lowest band is occupied ($n = 0$), and the Bloch functions that describe each one of the particles can be expanded in *Wannier functions* $W_{n,i}$, which are localized at each site, but aren't eigenstates of the single particle Hamiltonians, this is what we call the tight binding approximation. The whole expression for the Wannier functions is given by equation A.2, as well as the fact that they constitute a proper basis for our system:

$$W_R = \frac{1}{\sqrt{N}}\sum_k e^{-R\cdot k}\psi_k(x), \qquad \int W_{R'}^*(x)W_R(x)dx = \delta_{RR'}. \tag{A.2}$$

Where R is the lattice constant, k the quasi momentum whose values are located within the first Brillouin zone. Keeping in mind that $\hat{b}_i$ and $\hat{b}_i^\dagger$ are bosonic creation and annihilation operators, our field operators follow a similar commutation relation, as seen in equation A.3

$$\left[\hat{\Psi}(x), \hat{\Psi}^\dagger(x')\right] = \delta(x - x'), \qquad \left[\hat{\Psi}(x), \hat{\Psi}(x')\right] = \left[\hat{\Psi}^\dagger(x), \hat{\Psi}^\dagger(x')\right] = 0. \tag{A.3}$$

The calculations are straight forward, considering the approximations we've made to arrive here, but it's worth dividing the evaluation of the two terms that make the entire Hamiltonian, let's start with the first term on the right, where only single particle operators are present:

$$\begin{aligned}\hat{H}_0 &= \int dx \hat{\Psi}^\dagger(x)\left(-\frac{\hbar}{2m}\nabla_i^2 + \hat{V}_{ext}\right)\hat{\Psi}(x),\\ &= -\frac{\hbar}{2m}\int dx \hat{\Psi}^\dagger(x)\left(\nabla_i^2\right)\hat{\Psi}(x) + \int dx \hat{\Psi}^\dagger(x)\hat{V}_{ext}\hat{\Psi}(x).\end{aligned} \tag{A.4}$$

Fortunately enough, the process follows equally either for the first or the second term of $\hat{H}_0$ yields a result which we can factorize together into a single coefficient. Generally, the representation in second quantization of any single body operator is:

$$\begin{aligned}\hat{h}_i^{(2^{nd}Q)} &= \int dx \hat{\Psi}^\dagger(x)\hat{h}_l^{(1^{st}Q)}\hat{\Psi}(x),\\ &= \int dx \left(\sum_i W_i^*(x)\hat{b}_i^\dagger\right)\hat{h}_l\left(\sum_j W_j(x)\hat{b}_j\right) = \sum_{i,j}\hat{b}_i^\dagger\hat{b}_j\int dx W_i^*(x)\hat{h}_l W_j(x),\\ \hat{h}_i^{(2^{nd}Q)} &= \sum_{i,j} h_{ij}\hat{b}_i^\dagger\hat{b}_j,\end{aligned} \tag{A.5}$$

Now with equation A.5, we can go back to equation A.4 and replace it:

$$\begin{aligned}\hat{H}_0 &= -\frac{\hbar}{2m}\sum_{i,j} T_{ij}\hat{b}_i^\dagger\hat{b}_j + \sum_{i,j} V_{ij}\hat{b}_i^\dagger\hat{b}_j,\\ &= -\sum_{i,j}\left(\frac{\hbar}{2m}T_{ij} - V_{ij}\right)\hat{b}_i^\dagger\hat{b}_j = -\sum_{i,j} t_{ij}\hat{b}_i^\dagger\hat{b}_j,\\ &= -\sum_{i\neq j} t_{ij}b_i^\dagger\hat{b}_j - \sum_i t_{ii}\hat{b}_i^\dagger\hat{b}_i,\\ \hat{H}_0^{(2^{nd}Q)} &= -t\sum_{i\neq j}\hat{b}_i^\dagger\hat{b}_j - \mu\sum_i\hat{b}_i^\dagger\hat{b}_i.\end{aligned} \tag{A.6}$$

In an analogous way, the representation for two body operators in second quantization can be proved [65] to yield a really similar result as in equation A.5, but with four operators instead of two, accounting for the extra body, as shown in equation A.7:

$$\hat{V}^{(2^{nd}Q)} = \sum_{i,j,k,l}\langle ij|\,\hat{V}^{(1^{st}Q)}\,|kl\rangle\,\hat{b}_i^\dagger\hat{b}_j^\dagger\hat{b}_k\hat{b}_l. \tag{A.7}$$

As we are working in the *s-wave* approximation, where the internuclear interactions are way stronger than any other, therefore, the term that contributes the most energy is V_{nnnn} such that $V_{nnnn} \gg H_{ijkl}$ where $i \neq j \neq k \neq l$. This approximation is common practice for systems where particles interact via short-range potentials, as is our case with contact interactions given by equation A.8:

$$U(x,x') = \frac{4\pi\hbar^2 a}{m}\delta(x-x'). \tag{A.8}$$

A.2 Introduction of Spin

To be able to build the spin-1 Hamiltonian we have to go over the interaction term first, as it'll have to be expressed differently. A formalism which will yield useful is the addition of angular momenta [66], which takes two different basis into account, a coupled ($|S,M\rangle_c$) that considers the total spin of the two particles ($S = s_1 + s_2$) and their magnetic projections ($M = [-S,S]$) as well an uncoupled one ($|s_1,m_1,s_2,m_2\rangle = |m_1,m_2\rangle$), the local basis of each particle with their interactions is related to the coupled basis via the following relation,

$$\begin{aligned} |S,M\rangle_c &= \sum_{m_1,m_2} (|s_1,m_1\rangle \otimes |s_2,m_2\rangle)(\langle s_1,m_1| \otimes \langle s_2,m_2|)\,|S,M\rangle \\ &= \sum_{m_1,m_2} \langle s_1,m_1,s_2,m_2|S,M|s_1,m_1,s_2,m_2|S,M\rangle\,|s_1,m_1\rangle \otimes |s_2,m_2\rangle \\ &= \sum_{m_1,m_2} C_{m_1,m_2}\,|m_1,m_2\rangle\,, \end{aligned} \tag{A.9}$$

where C_{m_1,m_2} are the Clebsch-Gordon coefficients. In our case, since we are dealing with spin-1 particles, and as shown in equation A.8, the interactions occur through a contact pseudopotential, meaning particles interact only when they occupy the same lattice site.

Keeping in mind that we are interested in the dynamics at low energy levels, our main concern narrows down to the $s-$wave scattering, additionally our interaction channels are linear independent from one another, therefore the interaction can be decomposed as the sum of the projectors associated to each one of the processes ($\hat{P}_S$) with a total spin S, as shown in equation A.10:

$$\hat{V}_{\text{int}} = \sum_S \frac{4\pi\hbar a_S}{m}\hat{P}_S\,\delta(x-x') = \sum_S g_S\hat{P}_S\,\delta(x-x'), \tag{A.10}$$

where a_S and m correspond to the scattering lengths and the mass of the boson, respectively.

Symmetric States	**Antisymmetric States**	**Description**
$\|2,2\rangle_c = \|1,1\rangle$		Both spins in $m=+1$
$\|2,1\rangle_c = \frac{1}{\sqrt{2}}(\|1,0\rangle + \|0,1\rangle)$	$\|1,1\rangle_c = \frac{1}{\sqrt{2}}(\|1,0\rangle - \|0,1\rangle)$	Sym./antisym. combo of $m=1$ and $m=0$
$\|2,0\rangle_c = \frac{1}{\sqrt{6}}(\|1,-1\rangle + 2\|0,0\rangle + \|-1,1\rangle)$	$\|1,0\rangle_c = \frac{1}{\sqrt{2}}(\|1,-1\rangle - \|-1,1\rangle)$	Combo of opposite m and both in $m=0$
$\|2,-1\rangle_c = \frac{1}{\sqrt{2}}(\|-1,0\rangle + \|0,-1\rangle)$	$\|1,-1\rangle_c = \frac{1}{\sqrt{2}}(\|0,-1\rangle - \|-1,0\rangle)$	Sym./antisym. combo of $m=-1$ and $m=0$
$\|2,-2\rangle_c = \|-1,-1\rangle$		Both spins in $m=-1$
$\|0,0\rangle_c = \frac{1}{\sqrt{3}}(\|1,-1\rangle - \|0,0\rangle + \|-1,1\rangle)$		Singlet state for spin-1 bosons

Table A.1: Symmetric and antisymmetric two-particle states for spin-1 bosons in the coupled basis $|S, M\rangle$.

Drawing inspiration from equation 2.3, we can also write equation A.10 in second quantization as seen in equation A.11,

$$\hat{V}_{\text{int}} = \frac{1}{2}\sum_{i}\sum_{m_1,m_2}\sum_{m_1',m_2'}\sum_{S} g_S \langle m_1', m_2'| \hat{P}_S |m_1, m_2\rangle \hat{b}^{\dagger}_{i,m_2'}\hat{b}^{\dagger}_{i,m_1'}\hat{b}_{i,m_2}\hat{b}_{i,m_1}, \tag{A.11}$$

which can be expressed by doing the proper calculations of the projectors on each state. Every projector can be expressed as shown in equation A.12, and don't forget that our interest lies in the interaction channels that are symmetric, namely $S = 0, 2$

$$\hat{P}_S = \sum_{m=-S}^{S} |S, m\rangle_c \langle S, m|_c . \tag{A.12}$$

Alternatively to the expression shown in Eq. 2.6, the interaction term can also read as Eq. A.13, where the first term (U_0) is responsible for *charge scattering*, while the second (U_2) is responsible for *spin scattering* interactions.

$$\hat{H}_{\text{int}} = \frac{U_0}{2}\sum_{\sigma,i}\hat{n}_{\sigma,i}(\hat{n}_{\sigma,i} - 1) + \frac{U_2}{2}\sum_{\sigma,i}\left(\hat{\mathbf{F}}_i^2 - 2\hat{n}_{\sigma,i}\right) \tag{A.13}$$

where $\hat{\mathbf{F}}_i = \left(\hat{F}_i^x, \hat{F}_i^y, \hat{F}_i^z\right)$ is the total spin operator at site i, with

$$\hat{F}_i^\nu = \sum_{\sigma,\sigma'} \hat{b}_{i,\sigma}^\dagger \, [\hat{S}^\nu]_{\sigma,\sigma'} \, \hat{b}_{i,\sigma'}$$

being $[\hat{S}^\nu]_{\sigma,\sigma'}$ the matrix elements of the standard spin-1 matrices. $\hat{b}_{i,\sigma}^\dagger$ and $\hat{b}_{i,\sigma}$ are the creation and annihilation operators, respectively, which create and annihilate a particle with magnetic projection σ at site i. The interaction strength in the channel F is denoted as U_F and is defined as

$$U_F = c_F \int dx \, |W(x)|^4. \tag{A.14}$$

Appendix B

Appendix B

B.1 Construction of an MPO

When making the MPO representation [61] of operators which aren't local terms such as some examples shown in section 3.5 one has to approach it in a rigorous way, thinking of them as they really are, an operator acting in the i-th site, followed by either operators in $i+1$ or $i-1$, and identities everywhere else. One fundamental example corresponds to the operator seen in Eq. B.1

$$\hat{\mathcal{O}} = \sum_i \hat{\mathcal{B}}_i \hat{\mathcal{C}}_{i+1} + \hat{\mathcal{D}}_i \hat{\mathcal{E}}_{i+1} + \hat{\mathcal{A}}_i = \prod_{i=1}^{L-1} W^{[i]}, \tag{B.1}$$

Each and every operator that composes it, can be expressed as seen in B.2. While it might look messy at first, what is relevant to see what's left and what's right from the site of interest i, as it will determine where in the bulk matrix $W^{[i]}$

$$\begin{aligned}
\hat{\mathcal{B}}_{i-1}\hat{\mathcal{C}}_i &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{\mathcal{B}}_{i-1} \otimes \quad \hat{\mathcal{C}}_i \quad \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L, \\
\hat{\mathcal{B}}_{i}\hat{\mathcal{C}}_{i+1} &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{1}_{i-1} \otimes \quad \hat{\mathcal{B}}_i \quad \otimes \hat{\mathcal{C}}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L, \\
\hat{\mathcal{D}}_{i-1}\hat{\mathcal{E}}_i &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{\mathcal{D}}_{i-1} \otimes \quad \hat{\mathcal{E}}_i \quad \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L, \\
\hat{\mathcal{D}}_{i}\hat{\mathcal{E}}_{i+1} &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{1}_{i-1} \otimes \quad \hat{\mathcal{D}}_i \quad \otimes \hat{\mathcal{E}}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L, \\
\hat{\mathcal{A}}_i &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{1}_{i-1} \otimes \quad \hat{\mathcal{A}}_i \quad \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L, \\
\hat{1}_i &= \hat{1}_1 \otimes \hat{1}_2 \otimes \cdots \otimes \hat{1}_{i-1} \otimes \quad \hat{1}_i \quad \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_{L-1} \otimes \hat{1}_L
\end{aligned} \tag{B.2}$$

which can be summed up to Eq. B.3

$$\begin{array}{c|c|c|c|c}
\hat{1}_1 \otimes \cdots \otimes \hat{1}_{i-1} \otimes & \hat{1}_i & \hat{0} & \hat{0} & \hat{0} \\
\hline
\hat{1}_1 \otimes \cdots \otimes \hat{\mathcal{B}}_{i-1} \otimes & \hat{\mathcal{C}}_i & \hat{0} & \hat{0} & \hat{0} \\
\hline
\hat{1}_1 \otimes \cdots \otimes \hat{\mathcal{D}}_{i-1} \otimes & \hat{\mathcal{E}}_i & \hat{0} & \hat{0} & \hat{0} \\
\hline
\hat{1}_1 \otimes \cdots \otimes \hat{1}_{i-1} \otimes & \hat{\mathcal{A}}_i & \hat{\mathcal{B}}_i & \hat{\mathcal{D}}_i & \hat{1}_i \\
\hline
 & \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_L & \otimes \hat{\mathcal{C}}_{i+1} \otimes \cdots \otimes \hat{1}_L & \otimes \hat{\mathcal{E}}_{i+1} \otimes \cdots \otimes \hat{1}_L & \otimes \hat{1}_{i+1} \otimes \cdots \otimes \hat{1}_L
\end{array} \tag{B.3}$$

And to sum it up, see Eq. B.4.

$$W^{[0]} = \begin{bmatrix} \hat{0} & \hat{0} & \hat{0} & \hat{1} \end{bmatrix}; \quad W^{[i]} = \begin{bmatrix} \hat{1}_i & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{C}}_i & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{E}}_i & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{A}}_i & \hat{\mathcal{B}}_i & \hat{\mathcal{D}}_i & \hat{1}_i \end{bmatrix}; \quad W^{[L+1]} = \begin{bmatrix} \hat{1} \\ \hat{0} \\ \hat{0} \\ \hat{0} \end{bmatrix}. \tag{B.4}$$

At the end, with any MPO, if you'd like to construct a simple one without thinking much about it, and doesn't have terms which interact between non immediate neighbors

$$\begin{aligned} &= \begin{bmatrix} \hat{0} & \hat{0} & \hat{0} & \hat{1} \end{bmatrix} \cdots \begin{bmatrix} \hat{1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{C}}_{i-1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{E}}_{i-1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{A}}_{i-1} & \hat{\mathcal{B}}_{i-1} & \hat{\mathcal{D}}_{i-1} & \hat{1} \end{bmatrix} \begin{bmatrix} \hat{1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{C}}_i & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{E}}_i & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{A}}_i & \hat{\mathcal{B}}_i & \hat{\mathcal{D}}_i & \hat{1} \end{bmatrix} \begin{bmatrix} \hat{1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{C}}_{i+1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{E}}_{i+1} & \hat{0} & \hat{0} & \hat{0} \\ \hat{\mathcal{A}}_{i+1} & \hat{\mathcal{B}}_{i+1} & \hat{\mathcal{D}}_{i+1} & \hat{1} \end{bmatrix} \cdots \begin{bmatrix} \hat{1} \\ \hat{0} \\ \hat{0} \\ \hat{0} \end{bmatrix} \\ &= \cdots + \hat{\mathcal{B}}_{i-1}\hat{\mathcal{C}}_i + \hat{\mathcal{D}}_{i-1}\hat{\mathcal{E}}_i + \hat{\mathcal{B}}_i\hat{\mathcal{C}}_{i+1} + \hat{\mathcal{D}}_i\hat{\mathcal{E}}_{i+1} + \hat{\mathcal{A}}_{i-1} + \hat{\mathcal{A}}_i + \hat{\mathcal{A}}_{i+1} \cdots \end{aligned} \tag{B.5}$$

B.2 Numerical Basis

In accordance with the computational basis introduced in Chapter 4, the local Hilbert space for a single site contains the three spin states $\{-1, 0, 1\}$ together with a vacuum state $|\emptyset\rangle$. In matrix notation these basis vectors are represented as

$$|\emptyset\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad |-1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad |0\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

The vacuum simply enlarges the physical Hilbert space from dimension 3 to 4 without modifying the underlying SU(3) spin symmetry (see Sec. 2.5.2 of [67]), since $|\emptyset\rangle$ transforms as a singlet under spin rotations.

To describe on-site interactions, we introduce the extended basis obtained from the tensor product of two single-boson spaces. The resulting $4 \times 4 = 16$ states are

$$
\begin{array}{c|cccc}
\otimes & |\emptyset\rangle & |{-1}\rangle & |0\rangle & |1\rangle \\
\hline
|\emptyset\rangle & |\emptyset\,\emptyset\rangle & |{-1}\,\emptyset\rangle & |0\,\emptyset\rangle & |1\,\emptyset\rangle \\
|{-1}\rangle & |\emptyset\,{-1}\rangle & |{-1}\,{-1}\rangle & |0\,{-1}\rangle & |1\,{-1}\rangle \\
|0\rangle & |\emptyset\,0\rangle & |{-1}\,0\rangle & |0\,0\rangle & |1\,0\rangle \\
|1\rangle & |\emptyset\,1\rangle & |{-1}\,1\rangle & |0\,1\rangle & |1\,1\rangle
\end{array}
\tag{B.6}
$$

where $|a\,b\rangle \equiv |a\rangle \otimes |b\rangle$.

Since the spin operators act non-trivially only on the subspace $\{-1, 0, 1\}$, their 4×4 representation is obtained by padding the first row and column (corresponding to the vacuum) with zeros. Some of the operators used for correlation measurements in Chapter 4 are

$$
\begin{aligned}
\hat{S}^z &= |1\rangle\langle 1| - |{-1}\rangle\langle -1|\,, & \hat{n}_\sigma &= |\sigma\rangle\langle\sigma|\,, \\
\hat{S}^+ &= \sqrt{2}\big(|1\rangle\langle 0| + |0\rangle\langle -1|\big), & \hat{N} &= \sum_\sigma \hat{n}_\sigma, \\
\hat{S}^- &= \sqrt{2}\big(|0\rangle\langle 1| + |{-1}\rangle\langle 0|\big).
\end{aligned}
$$

Finally, to extend these single-particle operators to the two-boson Hilbert space (the maximum filling allowed in our model), we use the standard rule

$$
\hat{\mathcal{S}} = \hat{S} \otimes \mathbb{1} \;+\; \mathbb{1} \otimes \hat{S},
$$

which produces 16×16 operator matrices in the basis defined in Eq. (B.6).

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